习题二

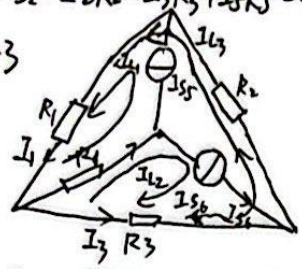
2-1

$$\begin{cases} I_1+I_3=I_{S_2} \\ -U_{S_1}+I_1R_1-U_{S_3}-I_3R_3=0 \end{cases} \Rightarrow I_3=0A$$

2-2

$$\left.\begin{array}{l} I_3=I_{S_1}+I_2 \\ I_5=I_3+I_{S_4} \\ -U_{S_2}+I_2R_2+I_3R_3+I_5R_5=0 \end{array}\right\} \Rightarrow I_3=1A$$

2-3



两个小圈为I_{L1}，I_{L2}，大圈为I_{L3}

回路电流法有

$I_{L1}=I_{S5}$ $I_{L2}=I_{S6}$ ①

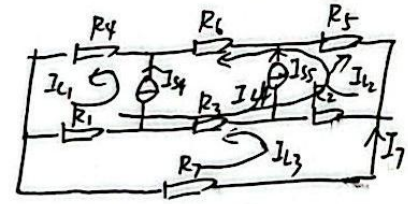
$R_1(I_{L1}+I_{L3})+R_3(I_{L3}-I_{L2})+R_2I_{L3}=0$ ②

$I_4=I_{L1}+I_{L2}$ ③

$I_2=I_{L3}$ $I_3=I_{L3}-I_{L2}$ $I_1=I_{L3}+I_{L1}$ ④

$\Rightarrow I_1=\frac{43}{6}A$ $I_2=\frac{13}{6}A$ $I_3=-\frac{23}{6}A$ $I_4=11A$

2-4



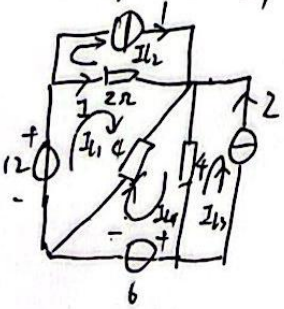
$I_{L1}=I_{S4}$ $I_{L2}=I_{S5}$

$$\begin{cases} I_{L3}R_7+(I_{L3}+I_{L2})R_2+(I_{L3}-I_{L4})R_3+(I_{L3}-I_{L1})R_1=0 \\ I_{L3}R_7+(I_{L4}-I_{L2})R_5+I_{L4}R_6+(I_{L1}+I_{L4})R_4=0 \end{cases}$$

$\Rightarrow I_{L3}=1A$ $I_{L4}=-\frac{1}{2}A$

$I_6=-I_{L4}=\frac{1}{2}A$ $I_7=I_{L3}=1A$

2-5



$I_{L2}=1A$ $I_{L3}=2A$

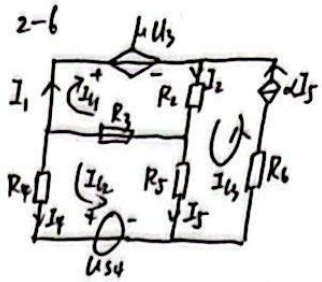
$-12+2(I_{L1}-I_{L2})+4(I_{L1}-I_{L4})=0$

$6+4(I_{L4}-I_{L1})+4(I_{L1}+I_{L3})=0$

$\Rightarrow I_{L1}=\frac{7}{4}A$ $I_{L4}=-\frac{3}{2}A$

$I_{L1}-I_{L2}=0.75A$

2-6



$(I_{L1}+I_{L2})R_3+\mu U_3+(I_{L3}+I_{L1})R_2=0$

$(I_{L2}+I_{L1})R_3+R_4I_{L2}+U_{S4}+(I_{L2}-I_{L3})R_5=0$

$I_{L3}=\alpha I_5$

$I_5=I_{L3}-I_{L2}$

$U_3=-R_3(I_{L1}+I_{L2})$

$\Rightarrow \begin{cases} I_{L1}=I_5 \\ I_{L2}=I_5 \\ I_{L3}=2I_5 \end{cases}$

$\Rightarrow I_5=0.8A$ $I_{L1}=0.8A$ $I_{L2}=0.8A$ $I_{L3}=1.6A$

$I_1=I_{L1}=0.8A$ $I_4=I_{L2}=0.8A$

2-7 （图小题的画了，对中间部分列KVL即可）

$I_{L1}=I_{S9}=9A$ $I_{L2}=I_{S7}=7A$ $I_{L3}=I_{S8}=8A$

$R_6(I_{L4}-I_{L1})+R_4(I_{L4}-I_{L2})+R_5(I_{L4}+I_{L3})=0$

$\Rightarrow I_{L4}=2.8A$

$I_1=I_{L2}-I_{L1}=-2A$ $I_2=-I_{L2}-I_{L3}=-15A$

$I_3=I_{L1}+I_{L3}=17A$ $I_4=I_{L4}-I_{L2}=-4.2A$

$I_5=I_{L3}+I_{L4}=10.8A$ $I_6=I_{L4}-I_{L1}=-6.2A$

2-8

对a点列节点电压法

$U_a(\frac{1}{60}+\frac{1}{30}+\frac{1}{20})=8/20+1.8+6/30$

$\Rightarrow U_a=24V$

$P_{IS}=I_S(U_a+I_5R_3)=108W$

$U_{R_3}=16V$ $I_{R_3}=0.8A$ $I_{R_6}=0.4A$

$I_{S_3}=-0.4A$

$P_{US_3}=-0.4A\cdot 8V=3.2W$

2-9

①点：$\frac{1}{4}(U_①-U_②)+(\frac{1}{5}+\frac{1}{20})(U_①-U_③)+\frac{1}{12}U_①=-\frac{10}{20}+3.5$

②点：$\frac{1}{4}(U_②-U_①)+\frac{1}{4}(U_②-U_③)+\frac{1}{4}U_②=0$

③点：$\frac{1}{4}(U_③-U_②)+(\frac{1}{5}+\frac{1}{20})(U_③-U_①)+\frac{1}{12}U_③=\frac{5}{6}+\frac{10}{20}$

$U_①=14V$ $U_②=\frac{26}{3}V$ $U_③=12V$

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2-10 (注意单独的电压源看作电流源)

设 $U_a = E_3$

左: $(U_b - U_a)\frac{1}{R_5} = I_{S4} - i$ (i为从左至右流入R_1的电流)

上点: $\frac{1}{R_6}U_c + \frac{1}{R_7}U_c = i + \frac{E_7}{R_7}$

$\beta I_7 = U_b - U_c$

$I_7 = \frac{E_7 - U_c}{R_7}$

2-11

左1: $U_a = U_{S1} = 1V$

中: $(U_b - U_a)\frac{1}{R_3} + U_b\frac{1}{R_5} + (U_b - U_c)\frac{1}{R_4} = 0 \Rightarrow U_b = 1.38V$

右一: $U_c = U_a + U_{S2} = 3V$

由此可求得各支路电流

$I_1 = -0.776A$ $I_2 = -0.903A$ $I_3 = 0.127A$

$I_4 = 0.403A$ $I_5 = 0.276A$

2-12

$U_a = 12V$

$U_b(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}) - \frac{1}{R_2}U_a - \frac{1}{R_3}U_c = 0$

$\frac{1}{R_3}(U_c - U_b) + \frac{1}{R_4}U_c = \alpha I_1 + I_5$

$\Rightarrow \begin{cases} U_a = 12V \\ U_b = 10V \\ U_c = 16V \end{cases}$

$I_1 = \frac{U_b}{R_1}$ $I_1 = 1A$

$I_2 = \frac{U_a - U_b}{R_2} = 0.4A$ $I_{US} = 0.8A + 2.8A = 3.2A$

$P_{US发} = 12 \times 3.2 = 38.4W$ $P_{IS} = 16W$

2-13

~~支路节点电压法~~ $U_a = U_b = 0$

对c点列节点电压法 以a点为参考点.

$U_c(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}) = \frac{U_{S1}}{R_1}$ $U_c = \frac{6}{11}V$

$I_3 = \frac{2}{11}A$ $I_5 = 1A$

$I_{ab} = I_3 - I_5 = -\frac{9}{11}A$

而 $U_3 = U_c = \frac{6}{11}V$

$\therefore q = -1.5$

2-14

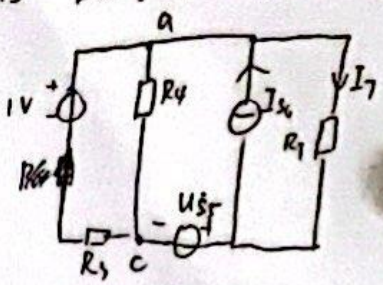
7Ω支路相当于短路/断路. 没有电流

$I = \frac{6V}{1 + (4\|4)} = 2A$

$4 = 3\|6 + 1 + 1$

$\therefore P_{US} = 6V \times 2A = 12W$.

2-15 首先对左侧并联的电压源进行化简



对A点节点电压法 c为参考点

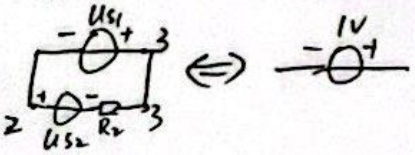
~~$\frac{1}{R_7}(U_a + \frac{1}{R_4 + R_5}U_a + \frac{1}{R_7}U_a =$~~

$(\frac{1}{R_4} + \frac{1}{R_5\|R_3})U_a + \frac{1}{R_7}(U_a - 5V) = \frac{1}{R_5\|R_3} + I_{S6}$

$\Rightarrow U_a = 9.7V$

$I_7 = \frac{U_a - 5}{R_7} = 0.67A$

注意等效



2-16

R_X 无电压 $\Rightarrow U_X = 0 \Rightarrow i_1 + i_4 = i_5$ ①

$i_4 = \frac{E_2}{R_4} = -1.5A$

$i_5 = \frac{U_5}{R_5}$ $U_5 = E_1 \times \frac{R_2\|R_3}{R_1 + R_2\|R_3}$ (R_1未知)

$U_5 = 1.5A \times (R_2\|R_3) = 20V$

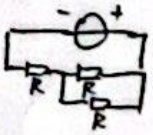
$i_5 = \frac{U_5}{R_5} = 1A$ 由① $\Rightarrow i_1 = 2.5A$

$R_1 = \frac{E_1 - 20V}{i_1} = 32\Omega$

(R_1未知. 为避免方程式复杂. 从右侧入手)

2-17

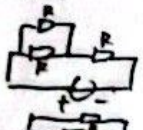
E_1单独作用时



$I_{11} = 1A$ $I_{21} = -0.5A$

$I_{31} = -0.5A$ $I_{41} = 1A$ $I_{51} = 0.5A$

E_2单独作用时

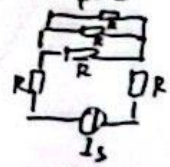


$I_{12} = 0.5A$ $I_{22} = 0.5A$

$I_{32} = -1A$ $I_{42} = -0.5A$

$I_{52} = 1A$

I_5单独作用时



$I_{13} = \frac{1}{5}A$ $I_{23} = \frac{1}{5}A$

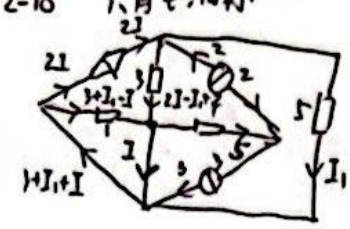
$I_{33} = \frac{4}{5}A$ $I_{43} = -\frac{2}{5}A$ $I_{53} = \frac{2}{5}A$

$\therefore I_1 = \sum I_{1i} = 1.83A$ $I_2 = \sum I_{2i} = 0.33A$

$I_3 = \sum I_{3i} = -1.167A$ $I_4 = \sum I_{4i} = 0.83A$ $I_5 = \sum I_{5i} = 2.167A$

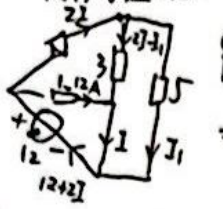
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2-18 只有电流源.



$\begin{cases} 5I_1 = 3(2I - I_1 + 2) \\ 3 + I_1 - I = 0 \end{cases} \Rightarrow \begin{cases} I' = 15A \\ I_1' = 12A \end{cases}$

只有电压源.



$\begin{cases} 5I_1 = 3(2I - I_1) \\ I + I_1 = 12 + 2I \end{cases} \Rightarrow \begin{cases} I'' = 48A \\ I_1'' = 36A \end{cases}$

$\therefore I_1 = I_1' + I_1'' = -24A$

2-19

$U = aU_S + bI_S + c$

$\begin{cases} 15 = 5a + b + c \\ 25 = -5a + b + c \\ 5 = -5a - b + c \end{cases} \Rightarrow \begin{cases} a = -1 \\ b = 10 \\ c = 10 \end{cases}$

$U_3 = 5a - b + c = -5V$

2-20

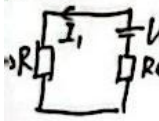
$I = aE_S + bI_S$

代入得 $a = 4$

$I' = 4 \times 3 = 12A$

2-21

将R右侧作戴维南等效



$I_1 = \frac{U_e}{aR + R_e}$

$\begin{cases} \frac{U_e}{R_e} = 2 \\ \frac{U_e}{R_e + 10} = 1.5 \end{cases} \Rightarrow \begin{cases} U_e = 60V \\ R_e = 30\Omega \end{cases}$

$\Rightarrow I_1'' = \frac{U_e}{R_e - 10} = 3A$

等左侧R变化等效成电流源.

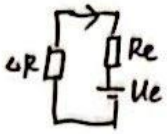
$I_2 = aI_1 + b$

$\begin{cases} \frac{1}{3} = 2a + b \\ 0.5 = 1.5a + b \end{cases} \Rightarrow \begin{cases} a = -\frac{1}{3} \\ b = 1 \end{cases}$

$I_2'' = -\frac{1}{3} \times 3 + 1 = 0A$

2-22

将R右侧戴维南等效



$I_1 = \frac{U_e}{R_e + aR}$

$\left.\begin{cases} \frac{U_e}{R_e} = 0.2 \\ \frac{U_e}{R_e + 10} = 0.16 \end{cases}\right\} \Rightarrow \begin{cases} U_e = 8V \\ R_e = 40\Omega \end{cases}$

$I_1'' = \frac{U_e}{R_e + 40} = 0.1A$

$U_2 = aI_1 + b$

$\left.\begin{cases} 5 = 0.2a + b \\ 6 = 0.16a + b \end{cases}\right\} \Rightarrow \begin{cases} a = -25 \\ b = 10 \end{cases}$

$U_2'' = -25 \times 0.1 + 10 = 7.5V$

2-23

逐级计算各电阻(两个回路)得 $U_{cd} = 18V$

$R_{cd} = R_1 // R_3 + R_2 // R_4 + R_5 = 12\Omega$

2-24

(1) 将R左侧戴维南等效

$R_{cd} = 3 + 8 // 56 = 10\Omega$

回路电流法 $\Rightarrow U_{cd} = 8V$



$P_{max} \Rightarrow R = R_{cd} = 10\Omega$

$P_{max} = \frac{1}{2} \times U_{cd} \times \frac{U_{cd}}{R_{cd} + R} = 1.6W$

(2) $R = 0$ 时 $I_{max} = \frac{U_{cd}}{R_{cd}} = 0.8A$

(3) $R \to \infty$ 时 $U_{max} = U_{cd} = 8V$

2-25

用回路(网孔)电流法测得 $U_{cd} = -8V$

计算 R_{cd} 只能用加压法.

在ab两端加电压 U_{ab}

$\Rightarrow R_{cd} = \frac{U_{ab}}{I_{ab}} = -5\Omega$

画出等效电路即可

2-26

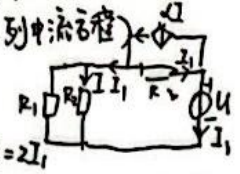
对R左侧作戴维南等效



求 U_{cd}. 计算每半上电压 $U_{cd} = 8 - 0.8R_3$

(即两个网孔到电流为零)

求 R_{cd}



$I = \frac{1}{5}I_1 \quad \alpha I = 2I_1$

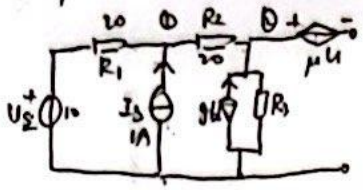
即 $(R_1 // R_2)I_1 - R_3 I_1 = U \Rightarrow R_{cd} = \frac{U}{I_1} = 20 - R_3$

$U_R = U_{cd} \cdot \frac{R}{R_{cd} + R} = (8 - 0.8R_3) \cdot \frac{R}{20 - R_3 + R}$

要使 U_R 不变, 则 $\begin{cases} 8 - 0.8R_3 = 0 \\ 20 - R_3 = 0 \end{cases}$ 或 $\Rightarrow \begin{cases} R_3 = 10\Omega \Rightarrow U = 0V \\ R_3 = 20\Omega \Rightarrow U = -8V \end{cases}$

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2-27



求 U_{oc} 或 I_{sc}：节点电压法

$\frac{1}{20}U_{①}+\frac{1}{20}(U_{①}-U_{②})=\frac{10}{20}+1$

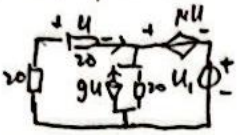
$\frac{1}{20}(U_{②}-U_{①})+\frac{1}{20}U_{②}=0.5U$

$U=U_{①}-U_{②}$

$\Rightarrow U_{①}=22V \quad U_{②}=14V \quad U=8V$

$\Rightarrow U_{oc}=U_{①}-\mu U=10V$

求 R_{eq}

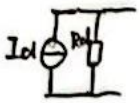


$U_1+\mu U+U+U=0 \Rightarrow U_1=-2.5U+gU$

~~$I_1=\left(\frac{U}{20}+\frac{2U}{20}\right)I_1=-\left(\frac{U}{20}+\frac{2U}{20}\right)=-\frac{3U}{20}$~~

$\Rightarrow R_{eq}=\frac{U_1}{I_1}=$ ~~16.67Ω~~ 13.33Ω

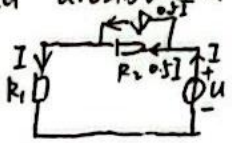
$I_{sc}=\frac{U_{oc}}{R_{eq}}=0.75A$



2-28

求 U_{oc} 断路 $U_{oc}=E=10V$

求 R_{eq} a.b端加压 U



$U=R_2\cdot 0.5I+R_1\cdot I$

$\Rightarrow R_{eq}=\frac{U}{I}=1500\Omega$

$I_{sc}=\frac{U_{oc}}{R_{eq}}=6.67\times10^{-3}A$

2-29

运用互易定理

将 4Ω 与 3Ω 电阻串，并进N网络

有 $I_1'+3I_2'\cdot(-6)=0$

$I_1U_1'+I_2U_2'=I_1'U_1+I_2'U_2$

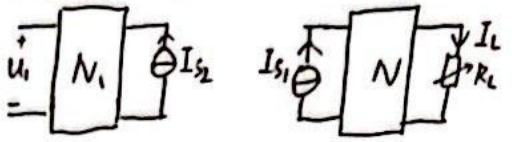
$U_1'=0 \quad I_2=0$ 故式子右侧为0

$\Rightarrow I_1'=1.8A$

$U_1=I_1'\cdot R=7.2V$

2-30（注意正负号）

由互易定理（将 R_1、R_2 并入N网络



$U_1I_{S1}+U_2\cdot(-I_L)=0+I_{S2}\cdot U_L$

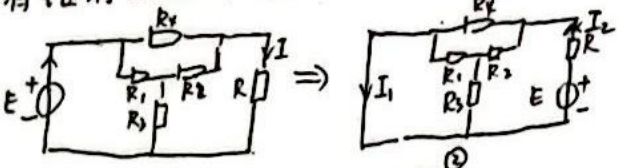
$\Rightarrow 2-2I_L=3R_LI_L \quad I_L=\frac{2}{3R_L+2}$

$P_L=I_L^2\cdot R_L=\frac{4R_L}{9R_L^2+12R_L+4}=\frac{4}{9R_L+\frac{4}{R_L}+12}$

$R_L=\frac{2}{3}\Omega$ 时 $P_{Lmax}=\frac{1}{6}W$

2-31

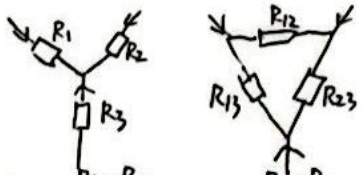
将电压源放在右侧，画出拓扑相同的电路



$-EI_1-IR\cdot I_2=0-EI-\cancel{0}RI_2I$ ①

$\Rightarrow$ 求出 I_1，I_2 即可

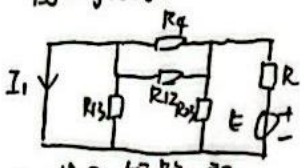
将 R_1、R_2、R_3 用 Y-Δ 变换



$R_1=\frac{R_{31}\cdot R_{12}}{R_{12}+R_{23}+R_{31}} \quad R_2=\frac{R_{12}\cdot R_{23}}{R_{12}+R_{23}+R_{31}} \quad R_3=\frac{R_{13}\cdot R_{23}}{R_{12}+R_{13}+R_{23}}$

$R_{12}=\frac{R_1R_2+R_2R_3+R_3R_1}{R_3} \quad R_{23}=\frac{R_1R_2+R_2R_3+R_3R_1}{R_1} \quad R_{31}=\frac{R_1R_2+R_2R_3+R_3R_1}{R_2}$

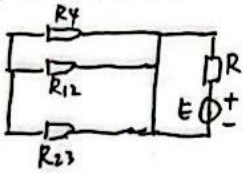
图二可化简为



$R_{12}=32\Omega \quad R_{23}=\frac{160}{3}\Omega$

$R_{31}=256\Omega$

I_1 将 R_{13} 短路，即



$R_4//R_{12}//R_{23}=10\Omega$

$\Rightarrow I_2=\frac{E}{R+R'}=2A$

~~$I_1=\frac{1}{2}I_2=1A$~~

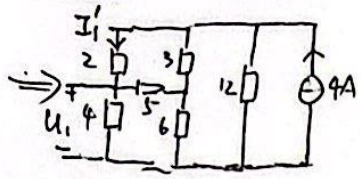
$I_1=I_2\times\frac{R_{23}}{R_{23}//R_{12}+R_{23}}=\frac{13}{8}A$

代入①式得 $I=I_1=1.625A$

CS 扫描全能王

2-32

互易定理



由互易定理 $4U = 4U_1 \Rightarrow U = U_1$

而左侧为平衡电桥

故为 6 11 9 1112 结构.

$9\|12 = \frac{36}{7}$

故 $I_1' = 4 \times \frac{\frac{36}{7}}{6+\frac{36}{7}} = \frac{24}{13}A$

$U_1 = I_1' \cdot 4 = 7.38V$

即 $U = U_1 = 7.38V$

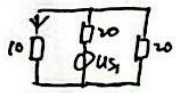
2-33 (互易与叠加综合) 电压源与电流源互换

仅 U_S 作用



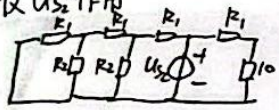
$I' = 0.8A$

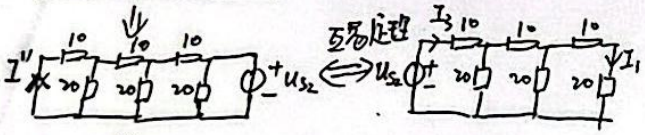
仅 U_{S_1} 作用



$I'' = \frac{U_{S_1}}{20+10\|20} \times \frac{20}{30} = -0.3A$

仅 U_{S_2} 作用

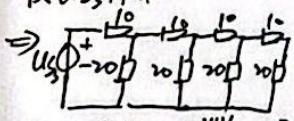




$I'' = -I_1$ $I_3 = \frac{24}{10+10} = 1.2A$ $I_1 = \frac{1}{8}I_3 = 0.15A$

$I'' = -0.15A$

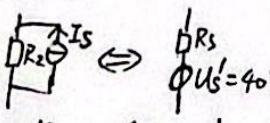
仅 U_{S_3} 作用



$I''' = \frac{-20}{10+10} \times \frac{1}{16} = -0.0625A$

仅 U_4 作用 $I'''' = \frac{-20}{10+10} \times \frac{1}{32} = \frac{-1}{32}A$

仅 I_S 作用



$I^{(6)} = -\frac{40}{10+10} \times \frac{1}{64} = -\frac{1}{32}A$

仅 U_{S_5} 作用

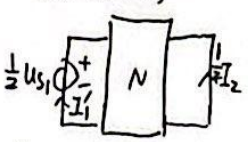
$I^{(7)} = -\frac{1}{64}A$

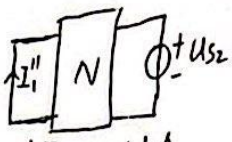
$\therefore I = \sum I^{(i)} = \frac{67}{320}A$

2-34 (注意判别方向)

不断使用互易定理与线性定理

以 U_{S_2} 为例





$\frac{1}{2}I_2 = \frac{1}{2}(5-3) = \frac{3}{2}A$ $I_1'' = -\frac{1}{2}I_2 = -1.5A$

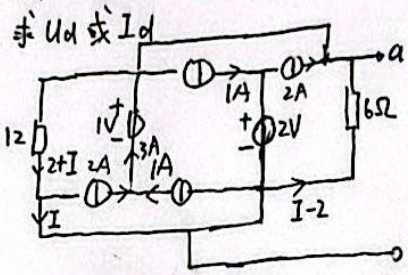
同理 $I_1''' = -6A$

$I_1 = I_1' + I_1'' + I_1''' = -2.5A$

2-35

$R_d = 12\|6 = 4\Omega$

求 U_{ol} 或 I_{ol}



即 $12(2+I) = -6(I-2) \Rightarrow I = -\frac{2}{3}A$

$U_{ab} = 12(2-\frac{2}{3}) = 16V$

$I_{ab} = \frac{U_{ab}}{R_d} = 4A$

2-36

<1> $I_5 = 0$

<2> $R_1 = 500\Omega$

$U_{oc} = \frac{2}{400+500} \cdot 400 - \frac{2}{800+800} \times 800 = -\frac{1}{9}V$

$R_{eq} = R_1\|R_2 + R_3\|R_4 = \frac{1600}{9}\Omega$ $I = \frac{U_{oc}}{R_{eq}+R_5} = -\frac{1}{12800}A$

2-37

使用网孔电流法

$I_{l_1}(20+30+50) + 50I_{l_2} - 30I_{l_3} = 0$

$50I_{l_1} + I_{l_2}(50+25+14) + 25I_{l_3} = 0$

$-30I_{l_1} + 25I_{l_2} + I_{l_3}(30+25+5) = -2$

$\Rightarrow I_{l_1} = -\frac{49}{1040}A$ $I_{l_2} = \frac{5}{104}A$ $I_{l_3} = -\frac{1}{13}A$

$I_5 = I_{l_1} + I_{l_2} = \frac{1}{1040}A = 9.61 \times 10^{-4}$

CS 扫描全能王

3-1

(1) $A=311$

初相位 $\varphi_0=\frac{\pi}{6}$

$\omega=2\pi f \Rightarrow f=50Hz$

$T=\frac{1}{f}=0.02s$

(2) 略

(3) $u_0=155.5V$

$u_{0.01s}=-269.7V$

3-2

$T=0.01s$

$\omega=\frac{2\pi}{T}=628rad/s$

$u=200\sin(628t-\frac{\pi}{3})$

3-3

$\omega=2\pi f=314rad/s$

$i=10\sqrt{2}\sin(314t+\frac{\pi}{2})$

3-4

(1) $u=100\sqrt{2}\sin(314t)$

$i=5\sqrt{2}\sin(314t-\frac{\pi}{2})$

(2) $i=5\sqrt{2}\sin 314t$

$u=5\sqrt{2}\sin(314t+\frac{\pi}{2})$

(3) $\frac{\pi}{2}$

3-5 $\dot{U}_1=220\angle-30°V$

$\dot{U}_2=220\angle30°V$

则 $\dot{I}=5\angle-90°V$.



3-6

$\dot{U}_1=220\angle70°$

$\dot{U}_2=200\angle30°$

$\dot{U}=\dot{U}_1+\dot{U}_2=270.4\angle23.35°$

3-7

$\dot{I}=10\angle60°$

$\dot{I}_1=5\angle30°$

$\dot{I}_2=\dot{I}-\dot{I}_1=6.196\angle83.79°$

3-8

$u_1=50\sqrt{2}\sin(314t+\frac{\pi}{6})$

$u_2=40\sqrt{2}\sin(314t-\frac{\pi}{3})$

$\dot{U}=\dot{U}_1+\dot{U}_2=64.03\angle-8.66°$

$u=64.03\sqrt{2}\sin(314t-8.66°)$

3-9 $P=\frac{U^2}{R}\Rightarrow U=\sqrt{PR}=300V$

3-10 $\dot{U}=220V$

$U_1=\frac{20}{20+30}\times220=88V$ $P_1=\frac{U_1^2}{R_1}=387.2W$

$U_2=132V$ $P_2=\frac{U_2^2}{R_2}=580.8W$

3-11

(1) $\dot{I}_L=1\angle0°$

$\dot{U}=j\omega L\dot{I}\Rightarrow\dot{U}=20\angle90°$

U的有效值为20V

(2) $u=20\sqrt{2}\sin(1000t+\frac{\pi}{2})$

$\frac{du}{dt}=20000\sqrt{2}\cos(1000t+\frac{\pi}{2})$

令 $1000t+\frac{\pi}{2}=\frac{\pi}{4}+2\pi\Rightarrow t=\frac{7\pi}{4000}s$

$i=20\sin\frac{7}{4}\pi=-1A$

$p=20\times(-1)=-20W$.

3-12

$X_L=2\pi f\cdot L=15.7\Omega$

$Z=R_L+jX_L=(10+j15.7)\Omega$.

$\dot{U}=20\angle0°$ $\dot{I}_L=\frac{\dot{U}}{Z}=11.81\angle-57.5°$

3-13

$X_L=2\pi fL=31.4\Omega$

$Z_L=j31.4+10=32.95\angle72.33°$

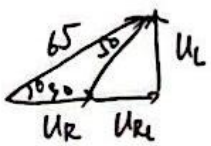
$\dot{U}=220\angle0°$ $\dot{I}=\frac{\dot{U}}{Z_L}=6.68\angle-72.33°$

$i=6.68\sqrt{2}\sin(314t-72.33°)$

$\dot{U}_R=\dot{U}\cdot\frac{R}{Z_L+R}=133.32\angle-72.33°\ 101.32\angle-46.3°$

$\dot{U}_L=166.94\angle-28.02°$ 化成瞬时值形式.

3-14 先画相量图



解三角形

$\cos\theta=\frac{15}{52}$

$U_L=48.07V$

$U_{RL}=13.75V$

$R_L=U_{RL}/\frac{U_R}{R}=0.4917\Omega$

$U_L=\frac{U_R}{R}\cdot jX_L\Rightarrow X_L=3.2\Omega$

$L=\frac{X_L}{2\pi f}=0.01H$

CS 扫描全能王

3-15

$\dot{U}_c = \frac{1}{j\omega C}\dot{I}_c$

$\dot{I}_c = j\omega C\dot{U}_c = 0.69\angle 90°$

$i_c = 0.69\sqrt{2}\sin(314t + \frac{\pi}{2})$

3-16

~~$Y = \frac{1}{j\omega C}$~~

$Y = j\omega C + \frac{1}{R} = 6.28\times10^{-3}j + 10^{-3}$

$\dot{U} = 500\angle 0°$

$\dot{I} = \dot{U}\cdot Y = 3.18\angle 80.95°$

$i = 3.18\sqrt{2}\sin(314t + 80.95°)$

3-17 串联支路

$\dot{U}_1 = 6\angle 0°$ $\dot{U}_2 = 10\angle 90°$ $\dot{U}_3 = 2\angle -90°$

$\dot{U} = \dot{U}_1 + \dot{U}_2 + \dot{U}_3 = 10\angle 53.13°$

~~$u = 10\sqrt{2}\sin($~~

V的读数为10V

3-18

$Z_{总} = Z_1 + Z_2 // Z_3 = 10\sqrt{29}\angle 68.2°$

$\dot{I}_1 = \frac{\dot{U}}{Z_{总}} = 4.09\angle -68.2°$

$\dot{I}_2 = \dot{I}_1 \times \frac{Z_3}{Z_2 + Z_3} = 3.65\angle -94.76°$

$\dot{I}_3 = 1.83\angle -4.77°$

3-19

$Y' = Y_2 // Z = Y_2 + \frac{1}{Z} = \sqrt{2}\angle -45°$

$Z_{ab} = \frac{1}{Y_1} + \frac{1}{Y'} = 1 + j2\ (\Omega)$

3-20

并联，电压相同，以R上电流电压为基准

$\dot{I}_R = 5\angle 0°$ $\dot{I}_L = 7\angle -90°$ $\dot{I}_C = 3\angle 90°$

$\dot{I} = \dot{I}_R + \dot{I}_L + \dot{I}_C = 6.40\angle -38.66°$

读数为6.4A 或 $\sqrt{41}$A

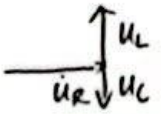
3-21 设 $\dot{I}_L = 10\angle 0°$ $\dot{I}_R = 10.0875\angle 90°$

$\dot{I} = \dot{I}_L + \dot{I}_R = 14.15\angle 45°$

$Z = \frac{1}{j\omega C} + \frac{R\cdot j\omega L}{R + j\omega L} = 2.83\angle -45°$

$\dot{U} = \dot{I}\cdot Z = 40V$

3-22



$\left.\begin{array}{l} U = \sqrt{U_R^2 + (U_L - U_C)^2} \\ U_{ab} = \sqrt{U_R^2 + U_L^2} \end{array}\right\} \Rightarrow \left\{\begin{array}{l} U_L = 131.25V \\ U_R = 72.62V \end{array}\right.$

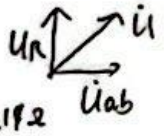
$\dot{U} = \dot{U}_R + \dot{U}_L + \dot{U}_C = 72.62\angle 0° + 131.25\angle 90° + 200\angle -90°$

$= 100\angle -43.43°$

$Z = \frac{\dot{U}}{\dot{I}} = 50\angle -43.53° = \frac{72.62 - 68.75j}{2}$

3-23 设 $\dot{I}_L = 10\angle 90°$ $\dot{I}_C = 15\angle 90°$

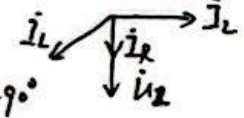
$\dot{I} = 5\angle 90°$ $\dot{U}_{ab} \to 0°$



$\dot{U}_R = 50\sqrt{2}\angle 90°$ $R = 10\sqrt{2}\Omega = 14.14\Omega$

$\dot{U}_{ab} = 50\sqrt{2}\angle 0°$ $X_L = 7.071\Omega$ $X_C = 4.714\Omega$

3-24 $\dot{U}$与$\dot{I}$同相



$\dot{I}_2 = 4\angle 0°$ $\dot{I}_R = 3\angle -90°$

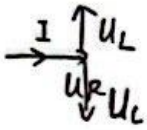
$\dot{I}_L = 5\angle -143.13°$ $R_L = 6\Omega$

~~$X_L = X_C = 11.25\Omega$~~ ~~$R_1 = \frac{X_C}{4}\times 3 = 9.375\Omega$~~

$\dot{U}_2 = 50\angle -90°$ $\dot{U}_R = 50\angle -90°$

$R = \frac{U_R}{I_R} = 16.67\Omega$.

3-25



(1) $|\dot{U}_R| = |\dot{U}_C - \dot{U}_L|$

$\frac{1}{\omega_1 C} - \omega_1 L = 1.5 \Rightarrow \omega_1 = 250\ rad/s$

(2) $\frac{1}{\omega_2 C} = \omega_2 L \Rightarrow \omega_2 = 500\ rad/s$

(3) $\omega_3 L - \frac{1}{\omega_3 C} = 1.5 \Rightarrow \omega_3 = 1000\ rad/s$

3-26



$\dot{I}_3 = 1\angle 90°$

$\dot{I}_1 = 0.5\angle 0°$

$\dot{I}_2 = 0.5\angle -90°$

$\dot{I}_4 = 0.5\angle 90°$

$Y = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{-jX_C} = 0.05 + j0.05\ (S)$

$Z = \frac{1}{Y} = 10 - j10\ (\Omega)$

CS 扫描全能王

3-27

$z_1 = R_1 - jX_{C1} = 30 - j20(\Omega)$

$z_2 = 50 - j100(\Omega)$

$z_3 = 100 + j50(\Omega)$

$z = z_1 + \frac{z_2 z_3}{z_2 + z_3} = 105 - j45(\Omega)$

$\dot{I} = \frac{\dot{U}}{z} = 1.75\angle 23.2°$

$\dot{I}_1 = \dot{I} \cdot \frac{z_3}{z_2 + z_3} = 1.24\angle 68.2°$

$\dot{I}_2 = \dot{I} \cdot \frac{z_2}{z_2 + z_3} = 1.24\angle -21.8°$

3-28

$z = z_1 + z_2 // z_3 = 69.78\angle 49.84°$

$\dot{I} = \frac{\dot{U}}{z} = 1.433\angle -49.84°$

$\dot{I}_1 = \dot{I} \cdot \frac{z_3}{z_2 + z_3} = 0.534\angle 13.59°$

$\dot{I}_2 = 1.28\angle -71.47°$

3-29

$z' = \frac{1}{Y_2} + z = 23.33 - j43.33(\Omega)$

$Y_{ab} = \frac{1}{z'} + Y_1 = \frac{13}{218} - \frac{7}{218}j(\Omega)$

$z_{ab} = \frac{1}{Y_{ab}} = 13 + j7(\Omega)$

3-30

用如压法计算. 设两端加压 $\dot{U}$

$\dot{U} = \dot{I}(-jX_C) + (\dot{I} + g\dot{U}_R) \cdot X_L + (1-\alpha)\dot{U}_R$ ①

而 $\dot{U}_R = (\dot{I} + g\dot{U}_R) \cdot R \Rightarrow \dot{U}_R = \frac{\dot{I}}{1 - gR}$ ②

②代入① 得

$$\frac{\dot{U}}{\dot{I}} = -jX_C + \frac{jX_C + R - \alpha R}{1 - gR}$$

3-31

~~$P = \frac{U^2}{R_L} = 4840W$~~ ×

$z_L = 10 + j15(\Omega)$ $\dot{U} = 220\angle 0°$

$\dot{I} = \frac{\dot{U}}{z_L} = 12.2\angle -56.3°$

$\tilde{S} = U \cdot I^* = P + jQ$

$\tilde{S} = U \cdot I^* = 1489.23 + j2233.85$

$P = 1489.23W$ $Q = 2233.85Var$.

3-32

$S = UI = 2200V \cdot A$

$S^2 = P^2 + Q^2 \Rightarrow Q = 1609.35Var$

$\cos\varphi = \frac{P}{S} = 0.6818$.

$P = I^2 R \Rightarrow R = 15\Omega$

$Q = I^2 X \Rightarrow X = 16.07\Omega$

3-33

a图电流从左向右. b图电流从右向左

故a图左发右吸. b图左吸右发

3-34.

(1) $S = U \cdot I = 2200V \cdot A$

$\cos\varphi = \frac{P}{S} = 0.5$

(2) $Q_{后} = \sqrt{S^2 - P^2} = \sqrt{\left(\frac{P}{\cos\varphi}\right)^2 - P^2} = 532.75Var$

$Q_{前} = \sqrt{S_1^2 - P^2} = 1905.26Var$

$\Delta Q = \frac{U^2}{-jX_C} \Rightarrow X_C = 35.26\Omega$

$X_C = \frac{1}{2\pi f C} \Rightarrow C = 90.26\mu F$

3-35



$I = \frac{P}{U} = 0.4A$ $R = 250\Omega$

$U_L = \sqrt{220^2 + 100^2} = 195.96V$ $X_L = \frac{U_L}{I} = 489.9\Omega$

$L = \frac{X_L}{2\pi f} = 1.56H$

一支灯: $P_1 = I^2 R = 40W$ $Q_1 = I^2 X_L = 78.38Var$

100支灯 $P = 100P_1 = 4kW$ $Q = 100Q_1 = 7.84kVar$

$\Delta Q = -Q = \frac{U^2}{-X_C} \Rightarrow X_C = 6.17\Omega = \frac{1}{2\pi f C}$

$\Rightarrow C = 516\mu F$

3-36

A发出功率 B吸收功率 C吸收功率

3-37

$\dot{I}_1 = \frac{\dot{U}_S}{3+j4} = 40\angle -53.13°A$ $\tilde{S} = \dot{U} \cdot \dot{I}_1^* = 4800 + j6400(VA)$

W_1的示数 $= P_1 = 4800W$.

$\dot{I}_2 = \frac{\dot{U}_S}{8 - j6} = 16 + j12(A)$ $\tilde{S} = \dot{U} \cdot \dot{I}_2^* = 3200 + (-j2400)V$

W_2的示数 $: -P_2 = -3200W$ (注意功率求电流流向)

$\dot{I} = \dot{I}_1 + \dot{I}_2 = 40 - j20(A)$ $\tilde{S} = \dot{U} \cdot \dot{I}^* = 8000 + j4000(V$

W的示数 $P_3 = 8000W$

3-38

$P_1 = I^2(R_1 + R_2) \Rightarrow R_1 + R_2 = 26\Omega$

$P_2 = I^2 R_2 \Rightarrow R_2 = 24.8\Omega$ $R_1 = 1.2\Omega$

$S_1 = U_1 \cdot I = 176V \cdot A$ $Q_1 = \sqrt{S_1^2 - P_1^2} = 141.99Var$

$S_2 = U_2 \cdot I = 160V \cdot A$ $Q_2 = \sqrt{S_2^2 - P_2^2} = 125.54Var$

$Q_1 = I^2(X_1 + X_L)$ $\Rightarrow \begin{cases} X_1 = 4.11\Omega \\ X_{L_2} = 31.38\Omega \end{cases}$

$Q_2 = I^2 X_L$

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3-39

a.b端作戴维南等效

$R_{eq} = Z_1 // Z_2 + Z_3 = 75 - j25(\Omega)$

$\dot{I} = \frac{\dot{U}_{S1} - \dot{U}_{S2}}{Z_1 + Z_2} = 0.457\angle -6.7^\circ$

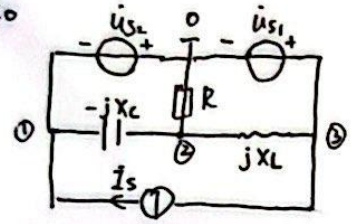
$\dot{U}_d = \dot{U}_{S1} - Z_1 \cdot \dot{I} = 96.73\angle -11.9^\circ$

要使功率最大，应共轭匹配。

即 $R = R_{eq}^* = 75 + j25(\Omega)$

$P_{max} = \frac{96.73^2}{2(75+75)} = 31.19W$

3-40



$\dot{U}_{①} = -\dot{U}_{S2}$ $\dot{U}_{③} = \dot{U}_{S1}$

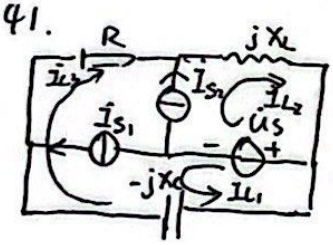
对②列节点电压方程

$(\dot{U}_{②} - \dot{U}_{①}) \cdot \frac{1}{-jX_C} + (\dot{U}_{②} - \dot{U}_{③}) \cdot \frac{1}{jX_L} + \dot{U}_{②} \cdot \frac{1}{R} = 0$

$\Rightarrow \dot{U}_{②} = 100\angle -150^\circ V$ $\dot{I}_R = \frac{\dot{U}_{②}}{R} = 2\angle -150^\circ A$

$\dot{I}_C = 3.86\angle -45^\circ A$ $\dot{I}_L = \frac{\dot{U}_{n2} - \dot{U}_{n3}}{jX_L} = 3.86\angle 105^\circ A$

3-41.



$\dot{I}_{L1} = \dot{I}_{S1}$ $\dot{I}_{L2} = \dot{I}_{S2}$

$-jX_C(\dot{I}_{L3} - \dot{I}_{L1}) + R \cdot \dot{I}_{L3} + jX_L(\dot{I}_{L2} + \dot{I}_{L3}) = 0$

$\dot{I}_{L3} = -2.91\angle 99.9^\circ A$

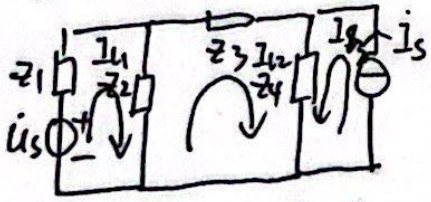
$\dot{I}_{XC} = \dot{I}_{L1} - \dot{I}_{L3} = 3.23\angle 62.4^\circ A$

$\dot{I}_R = \dot{I}_{L3} = -2.91\angle 99.9^\circ A$

$\dot{I}_L = \dot{I}_{L3} + \dot{I}_{L2} = 2.73\angle 120^\circ A$

3-42

用叠加定理或网孔电流法。



$-\dot{U}_S + Z_1\dot{I}_{L1} + Z_2(\dot{I}_{L1} - \dot{I}_{L2}) = 0$

$Z_2(\dot{I}_{L2} - \dot{I}_{L1}) + Z_3\dot{I}_{L2} + Z_4(\dot{I}_{L2} + \dot{I}_{L3}) = 0$

$\dot{I}_{L3} = \dot{I}_S = 5\angle -30^\circ$

$\Rightarrow$

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第四章

4-1

$\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} = 10^4 \text{ rad/s}$

$\rho = \sqrt{\frac{L}{C}} = 100\Omega \quad Q = \frac{\rho}{R} = 20$

$I = \frac{U}{R} = 4.8A$

$\dot{U}_L = \dot{I} \cdot j\omega_0 L = 480\angle 90° V$

$\dot{U}_C = 480\angle -90° V$

即 $U_L = U_C = 480V$

4-2

$\omega_0 L = \frac{1}{\omega_0 C} \quad (Y = \frac{1}{R} + j\frac{1}{\omega_0 L} + j\omega_0 C)$

$\Rightarrow \omega_0 = \sqrt{\frac{1}{LC}} = 1250 \text{ rad/s}$

$Q = \frac{\omega_0 C}{G} = \frac{1}{G\omega_0 L} = \frac{1}{G}\sqrt{\frac{C}{L}}$

$= 10$

$I_R = 2A \quad U_R = 100V$

$I_L = \frac{U_R}{\omega_0 L} = 20A = I_C$

4-3

$\omega = 1000 \text{ rad/s} = \sqrt{\frac{1}{LC}}$

所以发生并联谐振

相当于开路

即 $i = 0 \quad u_C = u = \sqrt{2} \times 100 \sin 1000t V$

4-4

并联谐振 $\omega_1 = \frac{1}{\sqrt{L_2 C}} \Rightarrow \boxed{L_2 = \frac{1}{\omega_1^2 C}}$

串联谐振：

$Z_{eq} = j\omega L_1 + \frac{j\omega L_2 \cdot \frac{1}{j\omega C}}{j\omega L_2 + \frac{1}{j\omega C}}$

$= j\omega L_1 + \frac{j\omega L_2}{1 - \omega^2 L_2 C} = 0$

即 $\omega_2 L_1 = \frac{\omega L_2}{\omega_2^2 L_2 C - 1}$

$\Rightarrow L_2 = \frac{L_1}{\omega_2^2 L_1 C - 1} =$

$\Rightarrow L_1 = \frac{L_2}{\omega_2^2 L_2 C - 1} = \boxed{\frac{1}{\omega_2^2 C - \omega_1^2 C}}$

4-5

$Y = \frac{1}{R + j\omega L} + j\omega C$

$= \frac{R}{R^2 + (\omega L)^2} + j\omega C - j\frac{\omega L}{R^2 + (\omega L)^2}$

即 $\omega C = \frac{\omega L}{R^2 + (\omega L)^2}$

$\Rightarrow \omega_0 = 1522.65 \text{ rad/s}$

$f_0 = \frac{\omega_0}{2\pi} = 242.34 Hz$

4-6

$Y = \frac{1}{R} + \frac{1}{j\omega L} + (1-\alpha) j\omega C \quad$ (加压法截准角)

即 $\frac{1}{\omega_0 L} = (1-\alpha)\omega_0 C \Rightarrow \omega_0 = \frac{1}{\sqrt{(1-\alpha)LC}}$

$Q = \frac{\sqrt{(1-\alpha)C}/L}{G} = \frac{R\sqrt{(1-\alpha)C}}{\sqrt{L}}$

4-7

1.2.3 同名. 通过观察法后可知

4-8

观察易知. 一个测的是 $U_1 + U_2$. 一个测的是 $|U_1 - U_2|$

即 $U_A > U_B$ 所以 1.2' 为同名端 1', 2 为同名端.

(写出各点电位即可验证).

4-9.

$Z_1 = R_1 + R_2 + j(\omega L_1 + \omega L_2 + 2\omega M)$

$Z_2 = R_1 + R_2 + j(\omega L_1 + \omega L_2 - 2\omega M)$

$P_1 = I_1^2 (R_1 + R_2) \Rightarrow R_1 + R_2 = 10\Omega$

$|Z_1| = |\frac{\dot{U}}{\dot{I}}| = 88\Omega$

$\omega L_1 + \omega L_2 + 2\omega M = \sqrt{88^2 - 10^2} = 87.43\Omega$ ①

$P_2 = I_2^2 (R_1 + R_2) \Rightarrow I_2 = 5A$

$|Z_2| = |\frac{\dot{U}}{\dot{I}}| = 44\Omega$

$\omega L_1 + \omega L_2 - 2\omega M = \sqrt{44^2 - 10^2} = 42.85\Omega$ ②

由 ① ② 知 $\omega M = 11.15\Omega \quad$ 而 $\omega = 314 \text{ rad/s}$

$M = 35.49mH$

4-10

图 a

$\dot{U} = \dot{I}_1 \cdot j\omega L_1 + \dot{I}_2 j\omega M$

$\dot{U} = \dot{I}_2 j\omega L_2 + \dot{I}_1 j\omega M$

$\Rightarrow \begin{cases} \dot{I}_1 = 2.73\angle -90° \\ \dot{I}_2 = 0.91\angle -90° \end{cases}$

$I = I_1 + I_2 = 3.64A \quad \dot{I} = 3.64\angle -90° A$

图 b

$\dot{U} = \dot{I}_1 j\omega L_1 - \dot{I}_2 j\omega M$

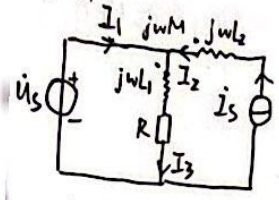
$\dot{U} = \dot{I}_2 j\omega L_2 - \dot{I}_1 j\omega M$

$\Rightarrow \begin{cases} \dot{I}_1 = 4.55\angle -90° \\ \dot{I}_2 = 2.73\angle -90° \end{cases}$

$\dot{I} = \dot{I}_1 + \dot{I}_2 = 7.28\angle -90° A$

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4-11



$(\dot{I}_1+\dot{I}_2)(jwL_1+R)-\dot{I}_2 jwM=\dot{U}_S$

$\dot{I}_2=\dot{I}_S=2\angle 90°A$

$\Rightarrow \dot{I}_1=4.06\angle -61.28°A$

$\tilde{S}=\dot{U}_S\cdot\dot{I}_1^*=39.02+j71.21(VA)$

$P_{US}=39.02W$

$\dot{U}_S+\dot{I}_2 jwL_2-(\dot{I}_1+\dot{I}_2)jwM=\dot{U}_{IS}$

$\dot{U}_{IS}=9.7\angle -23.72°$

~~$\tilde{S}=\dot{U}_{IS}$~~

$P=\dot{U}_{IS}\cdot I_S\cos\theta=-7.8W$

4-12

$\dot{U}_S=100\angle 0°$

$\dot{U}_S=(R_1+jwL_1)\dot{I}_1-\dot{I}_2\cdot jwM$

$\dot{U}_S=(R_2+jwL_2)\dot{I}_2-\dot{I}_1 jwM$

$\Rightarrow \begin{cases}\dot{I}_1=10.81\angle -19.57°\\ \dot{I}_2=8.26\angle -53.56°\end{cases}$

$P_1=U_S\cdot I_1\cos\theta_1=1018.55W$

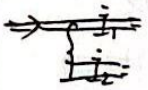
$P_2=U_S\cdot I_2\cos\theta_2=490.63W$

4-13

① 加压法 U_{ab}端加压 $\dot{U}_{ab}$

$\dot{U}_{ab}=(R_1+jX_{L_1})\dot{I}_1+jX_M\dot{I}_2+jX_{L_2}\dot{I}_2+jX_M\dot{I}_1$

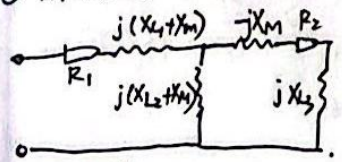
$\dot{U}_{ab}=(R_1+jX_{L_1})\dot{I}_1+jX_M\dot{I}_2+(\dot{I}_1-\dot{I}_2)(R_2+jX_{L_3})$



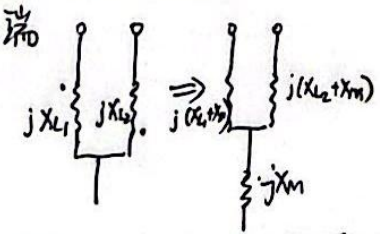
$\Rightarrow \dot{U}_{ab}=(16.04+j17.53)\dot{I}_1$

$\Rightarrow Z_{ab}=16.04+j17.53(\Omega)$

② 解耦法（更简单）



异名端



$Z=R_1+j(X_{L_1}+X_M)+j(X_{L_2}+X_M)//[R_2+j(X_{L_3}-X_M)]$

$=16.04+j17.53\Omega$.

4-14

$\begin{cases}\dot{U}_S=\dot{I}_1(R_1+jwL_1)+\dot{I}_2 jwM\\ 0=\dot{I}_2(R_2+Z+jwL_2)+\dot{I}_1 jwM\end{cases}$

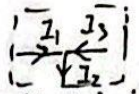
$\Rightarrow \begin{cases}\dot{I}_1=7.14\angle -74.53°\\ \dot{I}_2=1.65\angle 167.85°\end{cases}$

$P_1=U_S I_1\cos\varphi=418.83W$

$P_2=I_2^2\cdot R=136.13W$

$\eta=\frac{P_2}{P_1}=32.5\%$

4-15

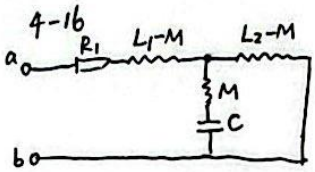


① $\dot{U}_{S_1}=\dot{I}_1\cdot jwL_1+\dot{I}_2 jwM_{12}+\dot{I}_2 jwL_2+\dot{I}_3 jwM_{12}+\dot{I}_3 jwM_{13}+R\dot{I}_1$

② $\dot{I}_3\cdot jwL_3+\dot{I}_1 jwM_{13}+\dot{I}_2 jwL_2+\dot{I}_3 jwL_1-j\frac{1}{wC}\dot{I}_3=\dot{U}_{S_2}$

③ $\dot{I}_1+\dot{I}_3=\dot{I}_2$

4-16



先解耦.

$Z_{ab}=R_1+jw(L_1-M)+(jwM+\frac{1}{jwC})//jw(L_2-M)$

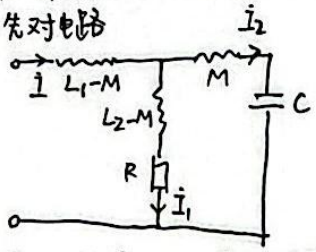
使虚部为0 $\Rightarrow X_C=280\Omega=\frac{1}{wC}$

$C\Rightarrow 3.57\times10^{-7}F$

4-17 ~~☆☆~~ (书后答案错误. 参见李玉玲60页)

例2-21

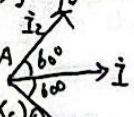
先对电路



$Z_{eq}=jw(L_1-M)+[R+jw(L_2-M)]//[jwM+\frac{1}{jwC}]$

<1> $X_C>X_M=wM$ 仍为容性

设 $\dot{I}=10\angle 0°A$ $\dot{I}_1=10\angle -60°A$ $\dot{I}_2=10\angle 60°A$



$\dot{I}_1[R+j(X_{L_2}-X_M)]=\dot{I}_2(jX_M-jX_C)$ ①

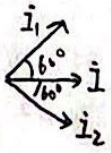
$\dot{U}=\dot{I}\cdot j(X_{L_1}-X_M)+\dot{I}_1\cdot[j(X_{L_2}-X_M)+R]$ ②

$P=I_1^2 R\Rightarrow R=4.33\Omega$

$\Rightarrow X_{L_2}=7.5\Omega$ $wM=5\Omega$ ①

由①可得 $X_{L_1}=10\Omega$或5Ω

<2> $X_C<X_M=wM$ 为感性



$\dot{I}_1=10\angle 60°A$ $\dot{I}_2=10\angle -60°A$

$\Rightarrow X_{L_2}=12.5\Omega$ $wM=15\Omega$

而 $X_{L_1}=10\Omega$或15Ω 因为 $M\le\sqrt{L_1L_2}$

所以<2>应舍去

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4-18

$R'=(\frac{N_1}{N_2})^2 R$

$\Rightarrow N_1:N_2=4:1$

4-19

在ab端加压 U_{ba}

$U_b=kU_a$

$I_a=kI_b$

~~易知 $U_a=R(I_a+I_b)=kU_a$~~

~~即 $R(k+1)$~~

易知 $I_R=I_a-I_b=(k-1)I_b$

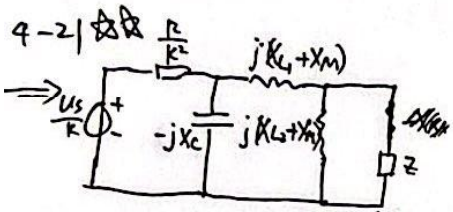
即 $R(k-1)I_b=U_a$

$\therefore \frac{U_b-U_a}{I_b}=\frac{(k-1)U_a}{I_b}=R(k-1)^2=R(\frac{N_1}{N_2}-1)^2$

4-20

$U_{eq}=\frac{N_2}{N_1}U_S \quad Z_{eq}=(\frac{N_2}{N_1})^2 Z$

4-21



将 Z 与 X_M 左侧戴维南等效

得到 Z_{eq}、U_{eq}

其中 $Z_{eq}-X_M=Z^*=80-j60$

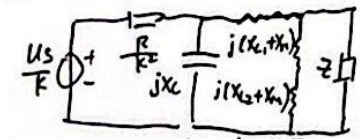
$Z_{eq}=\frac{R}{k^2}\,//\,-jX_C\,//\,j(X_{L_1}+X_{L_2}+2X_M)$

$\Rightarrow \frac{k^2}{5}+j\frac{1}{X_C}+\frac{1}{j500}=\frac{1}{80-j60}$

$\Rightarrow$ 实部、虚部分别相等

$k=0.2=\frac{N_1}{N_2} \quad X_C=125\Omega$

上面的图应当画成



即 $U_{oc}=\frac{U_S}{K}\cdot\frac{jX_C\,//\,[j(X_{L_1}+X_M)+j(X_{L_2}+X_M)]}{\frac{R}{k^2}+[j(X_{L_1}+X_M)+j(X_{L_2}+X_M)]\,//\,jX_C}$

$[j(X_{L_1}+X_M)+j(X_{L_2}+X_M)]\,//\,jX_C=-\frac{500}{3}j$

代入得 $U_{oc}=40\angle -36.87°$

$P_{max}=\frac{40^2}{4\times 80}=5W$

4-22

先将电路作Δ-Y转换

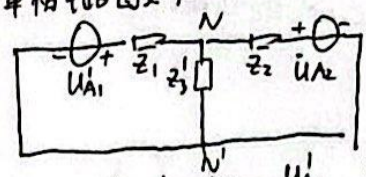
电源作Δ-Y转换

$\dot{U}'_{A_1}=\dot{U}_{A_1}/\sqrt{3}\angle -30°=220\angle -30°V$

电阻作Δ-Y转换

$Z'_3=Z_3/3=8+j6(\Omega)$

单相电路图如下



$U_{NN'}(\frac{1}{Z_1}+\frac{1}{Z_2}+\frac{1}{Z'_3})=\frac{U'_{A_1}}{Z_1}+\frac{U_{A_2}}{Z_2}$

$\Rightarrow U_{NN'}=52.23\angle -60.45° \quad I'_{NN}=5.223\angle -97.32°$

$\therefore I_O=I'_{NN}/\sqrt{3}\angle 30°=3.02\angle -67.32°$

$I_{Z_1}=\frac{U_N-U_{N'}}{Z_1}=8.848\angle -111.4°$

$I_{相}=I_{线}/\sqrt{3}\angle 30°=5.11\angle -81.4° \quad P=3U_{相}I_{相}\cos\theta=871W$

4-23

将 Z_Δ 作 Y-Δ 变换，即 $Z_Y=\frac{1}{3}Z_\Delta=(15-j5\sqrt{3})\Omega$

单相电路：



即 $\dot{I}_{A_1}=10\angle -30°$

$\dot{I}_{Z_Y}=10\angle 30°$

$\dot{I}_{AB}=\dot{I}_{Z_Y}/\sqrt{3}\angle 30°=5.77\angle 60°$

$\tilde{S}=\dot{I}^2_{Z_Y}\cdot Z_Y=1500-j866(VA)$

$P=3P_1=3\times 1500=4500(W)$

4-24

对 Z_2 作 Y-Δ 变换，即 $Z'_2=\frac{1}{3}Z_2=(6-j7)\Omega$



$\dot{I}_1=\frac{U_A}{Z_1+Z_2}=10\angle 36.87°$

$\dot{I}_{Z_\Delta}=\dot{I}_1/\sqrt{3}\angle 30°=5.77\angle 66.87°$

$\dot{I}_2=-\dot{I}_{Z_\Delta}\cdot\angle 120°=5.77\angle 6.87°$

$\dot{U}=\dot{I}_{Z_\Delta}\cdot Z_2=159.59\angle 17.47°V$

4-25

设 $\dot{U}_A=220\angle 0°V \quad \dot{U}_B=220\angle -120° \quad \dot{U}_C=220\angle 120°V$

$\dot{I}_A=\frac{U_A}{Z}=22\angle -36.87° \quad \dot{I}_B=22\angle -156.87°$

$\dot{U}_{AC}=\dot{U}_A-\dot{U}_C=381\angle -30°$

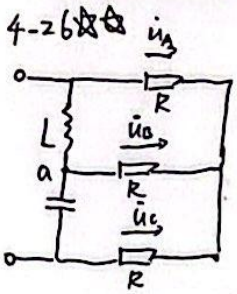
$P_1=U_{AC}\cdot I_a\cos(36.87-30°)=8322W$

$\dot{U}_{BC}=\dot{U}_B-\dot{U}_C=381\angle -90°$

$P_2=U_{BC}\cdot I_B\cos(156.87°-90°)=3293W$

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4-26☆☆



设 $\dot{u}_A = U_1\angle 0°$ $\dot{u}_B = U_1\angle -120°$ $\dot{u}_C = U_1\angle 120°$

对a点作KCL得

$$\frac{\dot{u}_A-\dot{u}_B}{j\omega L} = \frac{\dot{u}_B-\dot{u}_C}{\frac{1}{j\omega C}} + \frac{\dot{u}_B}{R}$$

$$\Rightarrow R = \frac{\omega L}{\sqrt{3}} = \frac{1}{\sqrt{3}\omega C}$$

$$L = \frac{\sqrt{3}R}{\omega} = 0.055H \quad C = \frac{1}{\sqrt{3}\omega R} = 1.84\times10^{-4}F$$

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习题五 双口网络

5-1

$$\begin{cases}\dot{U}_1 = r_{m1}\dot{I}_2 + (\dot{I}_1+\dot{I}_2)R \\ \dot{U}_2 = r_{m2}\dot{I}_1 + (\dot{I}_2+\dot{I}_1)R\end{cases}$$

$$\Rightarrow \begin{cases}\dot{U}_1 = R\dot{I}_1 + (r_{m1}+R)\dot{I}_2 \\ \dot{U}_2 = (r_{m2}+R)\dot{I}_1 + R\dot{I}_2\end{cases}$$

$$Z = \begin{bmatrix} R & r_{m1}+R \\ r_{m2}+R & R\end{bmatrix}$$

5-2

① $\dot{U}_1 = \dot{I}_1\cdot j\omega L_1 + j\omega M\dot{I}_2$

$\dot{U}_2 = \dot{I}_2 j\omega L_2 + j\omega M\dot{I}_1$

$$\Rightarrow Z = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2\end{bmatrix}$$

② $Z_{11} = \left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{I}_2=0} = j\omega L_1$

$Z_{12} = \left.\frac{\dot{U}_1}{\dot{I}_2}\right|_{\dot{I}_1=0} = j\omega M$

$Z_{21} = \left.\frac{\dot{U}_2}{\dot{I}_1}\right|_{\dot{I}_2=0} = j\omega M$

$Z_{22} = \left.\frac{\dot{U}_2}{\dot{I}_2}\right|_{\dot{I}_1=0} = j\omega L_2$

5-3

$\dot{U}_1 = \dot{I}_1(R_1+R_2) + (\dot{I}_1+\dot{I}_2)\frac{1}{j\omega C}$

$\dot{U}_2 = \dot{I}_2\cdot j\omega L + (\dot{I}_1+\dot{I}_2)\frac{1}{j\omega C}$

$\Rightarrow \dot{U}_1 = (30-j40)\dot{I}_1 - j40\dot{I}_2$

$\dot{U}_2 = -j40\dot{I}_1 + j10\dot{I}_2$

$$Z = \begin{bmatrix} 30-j40 & -j40 \\ -j40 & j10\end{bmatrix}$$

5-4

$\dot{I}_C = \dot{U}_1\cdot j\omega C \quad \dot{I}_R = \dot{U}_2\cdot\frac{1}{R}$

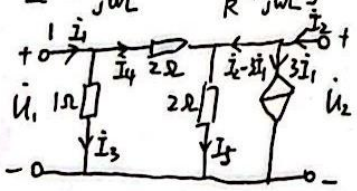
$\dot{I}_L = (\dot{U}_1-\dot{U}_2)\cdot\frac{1}{j\omega L}$

$\dot{I}_1 = \dot{I}_C + \dot{I}_L = (j\omega C + \frac{1}{j\omega L})\dot{U}_1 - \frac{1}{j\omega L}\dot{U}_2$

$\dot{I}_2 = \dot{I}_R - \dot{I}_L = -\frac{1}{j\omega L}\dot{U}_1 + (\frac{1}{R}+\frac{1}{j\omega L})\dot{U}_2$

即 $Y = \begin{bmatrix} j\omega C+\frac{1}{j\omega L} & -\frac{1}{j\omega L} \\ -\frac{1}{j\omega L} & \frac{1}{R}+\frac{1}{j\omega L}\end{bmatrix}$

5-5



$\dot{I}_3 = \dot{U}_1 \quad \dot{I}_4 = \frac{1}{2}(\dot{U}_1-\dot{U}_2)$

$\dot{I}_1 = \dot{I}_3+\dot{I}_4 = \frac{3}{2}\dot{U}_1 - \frac{1}{2}\dot{U}_2$

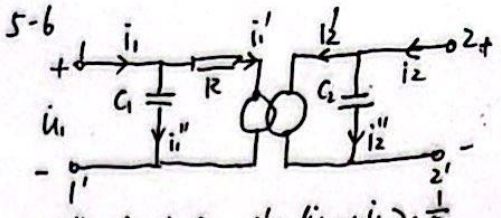
$\dot{I}_5 = \frac{1}{2}\dot{U}_2$

$\dot{I}_2 - 3\dot{I}_1 = -\dot{I}_4 + \dot{I}_5$

$\dot{I}_2 = 4\dot{U}_1 - \frac{1}{2}\dot{U}_2$

$\therefore Y = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ 4 & -\frac{1}{2}\end{bmatrix}$

5-6



$\dot{I}_1'' = \dot{U}_1\cdot j\omega C_1 \quad \dot{I}_1' = (\dot{U}_1 - n\dot{U}_2)\cdot\frac{1}{R}$

$\dot{I}_2' = n\dot{I}_1' = \frac{-n}{R}(\dot{U}_1 - n\dot{U}_2) \quad \dot{I}_2'' = \dot{U}_2\cdot j\omega C_2$

$\dot{I}_1 = \dot{I}_1' + \dot{I}_1'' = (\frac{1}{R}+j\omega C_1)\dot{U}_1 - \frac{n}{R}\dot{U}_2$

$\dot{I}_2 = \dot{I}_2' + \dot{I}_2'' = -\frac{n}{R}\dot{U}_1 + (\frac{n^2}{R}+j\omega C_2)\dot{U}_2$

$$Y = \begin{bmatrix} \frac{1}{R}+j\omega C_1 & -\frac{n}{R} \\ -\frac{n}{R} & \frac{n^2}{R}+j\omega C_2\end{bmatrix}$$

5-7 T的参数矩阵　　H参数矩阵（不考）

$$\begin{cases}\dot{U}_1 = A\dot{U}_2 + B(-\dot{I}_2) \\ \dot{I}_1 = C\dot{U}_2 + D(-\dot{I}_2)\end{cases} \qquad \begin{cases}\dot{U}_1 = H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ G\dot{I}_2 = H_{21}\dot{I}_1 + H_{22}\dot{U}_2\end{cases}$$

$\dot{U}_1 = Z_1\cdot\dot{I}_1 \quad \dot{U}_2 = (\dot{I}_2 + G\dot{U}_1)Z_2$

$$H = \begin{bmatrix} Z_2 & \cancel{GZ_1D} \\ -GZ_1 & \frac{1}{Z_2}\end{bmatrix}$$

5-8 由上述H参数矩阵

$H_{11} = \left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{U}_2=0} = Z \quad H_{21} = \left.\frac{\dot{I}_2}{\dot{I}_1}\right|_{\dot{U}_2=0} = \frac{N_1}{N_2}$

$H_{12} = \left.\frac{\dot{U}_1}{\dot{U}_2}\right|_{\dot{I}_1=0} = -\frac{N_1}{N_2} \quad H_{22} = \left.\frac{\dot{I}_2}{\dot{U}_2}\right|_{\dot{I}_1=0} = 0$

$$H = \begin{bmatrix} Z & \frac{N_1}{N_2} \\ -\frac{N_1}{N_2} & 0\end{bmatrix}$$

5-9 $\dot{I}_1 = \dot{U}_1\cdot\frac{1}{Z_1} + 3\dot{I}$

$\dot{I}_2 = -3\dot{I} + \dot{I} = -2\dot{I}$

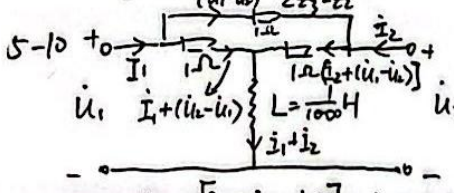
$\dot{U}_2 = -2\dot{I}\cdot Z_3 + \dot{I}\cdot Z_2$

$\Rightarrow \dot{U}_1 = Z_1\dot{I}_1 + \frac{-3Z_1}{2Z_3-Z_2}\dot{U}_2$

$\dot{I}_2 = 0\,\dot{I}_1 + \frac{2}{2Z_3-Z_2}\dot{U}_2$

$$\therefore H = \begin{bmatrix} Z_1 & \frac{-3Z_1}{2Z_3-Z_2} \\ 0 & \frac{2}{2Z_3-Z_2}\end{bmatrix}$$

5-10



$j\omega L = j0.314$

左：$\dot{U}_1 = 1[\dot{I}_1+(\dot{U}_2-\dot{U}_1)] + j0.314(\dot{I}_1+\dot{I}_2)$

$\dot{U}_2 = 1[\dot{I}_2+(\dot{U}_1-\dot{U}_2)] + j0.314(\dot{I}_1+\dot{I}_2)$

消路1 $\Rightarrow \dot{U}_1 = (1.53-j0.5)\dot{U}_2 + (0.84-j0.166)(-\dot{I}_2)$

而对称电路 $A=D=1.53-j0.5$ ~~$B=C=0.84-j$~~

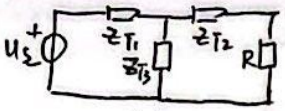
$B=0.84-j0.166$ 由 $AD-BC=1 \Rightarrow C=1.59-j1.5$

$$\therefore T = \begin{bmatrix} 1.53-j0.5 & 0.84-j0.166 \\ 1.59-j1.5 & 1.53-j0.5\end{bmatrix}$$

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5-11

画出T型等效电路



$Z_{T1}=\frac{A-1}{C}=3\Omega$ $Z_{T2}=\frac{D-1}{C}=1.2\Omega$

$Z_{T3}=\frac{1}{C}=2\Omega$

对R左端作戴维南等效

$Z_{eq}=Z_{T1}//Z_{T3}+Z_{T2}=2.4\Omega$

$R=Z_{eq}=2.4\Omega$ 时负载获最大功率

~~$P_{max}=\frac{U_S^2}{4R}=8.4375W$~~

~~$P_{US}=2P_{max}=16.875W$~~

先求 U_{eq} $U_{eq}=U_S\cdot\frac{Z_{T3}}{Z_{T3}+Z_{T1}}=3.6V$

$P_{max}=\frac{U_{eq}^2}{4R}=1.35W$

$I_R=\frac{U_{eq}}{2R}=0.75A$

$I_S=I_R/\frac{Z_{T3}}{Z_{T2}+R+Z_{T3}}=2.1A$

$P_{US}=I_S U_S=18.9W$

5-12 (易算错)

① $\dot U_1=\dot I_1\cdot\frac{1}{j\omega C}+(\dot I_1+\dot I_2)\cdot j\omega L$

$\dot U_2=\dot I_2\cdot\frac{1}{j\omega C}+(\dot I_1+\dot I_2)j\omega L$

$\Rightarrow \frac{\frac{1}{j\omega C}+j\omega L}{j\omega L}\dot U_2-\dot U_1=[\frac{(\frac{1}{j\omega C}+j\omega L)^2}{j\omega L}-j\omega L]\dot I_2$

即 $A=1-\frac{1}{\omega^2}$ $B=j\cdot\frac{1-2\omega^2}{\omega^3}$

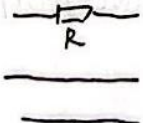
对称网络 故 $D=A=1-\frac{1}{\omega^2}$

无受控源 $AD-BC=1$

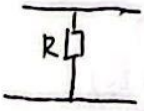
$\Rightarrow C=\frac{1}{j\omega}$

故T型等效 $T=\begin{bmatrix}1-\frac{1}{\omega^2} & j\frac{1-2\omega^2}{\omega^3}\\ \frac{1}{j\omega} & 1-\frac{1}{\omega^2}\end{bmatrix}$

② 首先有一种等效各类结构模型需注意



$T=\begin{bmatrix}1 & R\\ 0 & 1\end{bmatrix}$



$T=\begin{bmatrix}1 & 0\\ \frac{1}{R} & 1\end{bmatrix}$

因此可以看作三个T矩阵级联

$T_3=T_1=\begin{bmatrix}1 & \frac{1}{j\omega C}\\ 0 & 1\end{bmatrix}$ $T_2=\begin{bmatrix}1 & 0\\ \frac{1}{j\omega L} & 1\end{bmatrix}$

$T=T_1T_2T_3=\begin{bmatrix}1 & \frac{1}{j\omega}\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ \frac{1}{j\omega} & 1\end{bmatrix}\begin{bmatrix}1 & \frac{1}{j\omega}\\ 0 & 1\end{bmatrix}$

$=\begin{bmatrix}1-\frac{1}{\omega^2} & \frac{1-2\omega^2}{\omega^3}j\\ \frac{1}{j\omega} & 1-\frac{1}{\omega^2}\end{bmatrix}$

5-13 逆传输矩阵 T' 有

$\begin{cases}\dot U_2=A'\dot U_1+B'(-\dot I_1)\\ \dot I_2=C'\dot U_1+D'(-\dot I_1)\end{cases}$

根据电路可以写出

$\dot U_1=(\dot I_1+Y_1\dot U_2)Z_1$

$\dot U_2=(\dot I_2-Y_2\dot U_1)Z_2$

整理可得 $\begin{cases}\dot U_2=\frac{1}{Y_1Z_1}\dot U_1+\frac{1}{Y_1Z_1}(-\dot I_1)\\ \dot I_2=Y_2+\frac{1}{Z_1Z_2Y_1}\dot U_1+\frac{1}{Y_1Z_2}(-\dot I_1)\end{cases}$

即 $T'=\begin{bmatrix}\frac{1}{Y_1Z_1} & \frac{1}{Y_1}\\ Y_2+\frac{1}{Y_1Z_1Z_2} & \frac{1}{Y_1Z_2}\end{bmatrix}$

(答案中 B'、C' 未取负号，在于对逆传输参数定义不明)

5-14 (H矩阵不考，不计算)

A图Π型：

$\dot I_{Z1}=\dot U_1\cdot\frac{1}{Z_1}$ $\dot I_{Z3}=(\dot U_1-\dot U_2)\cdot\frac{1}{Z_3}$

$\dot I_{Z2}=\dot U_2\cdot\frac{1}{Z_2}$

$\begin{cases}\dot I_1=\dot U_1\cdot\frac{1}{Z_1}+(\dot U_1-\dot U_2)\cdot\frac{1}{Z_3}\\ \dot I_2=\dot U_2\cdot\frac{1}{Z_2}-(\dot U_1-\dot U_2)\cdot\frac{1}{Z_3}\end{cases}$ 通过化简方程即可得到各参数

$Z=\begin{bmatrix}\frac{Z_1(Z_2+Z_3)}{Z_1+Z_2+Z_3} & \frac{Z_1Z_2}{Z_1+Z_2+Z_3}\\ \frac{Z_1Z_2}{Z_1+Z_2+Z_3} & \frac{Z_2(Z_1+Z_3)}{Z_1+Z_2+Z_3}\end{bmatrix}$ $Y=\begin{bmatrix}\frac{1}{Z_1}+\frac{1}{Z_3} & -\frac{1}{Z_3}\\ -\frac{1}{Z_3} & \frac{1}{Z_3}+\frac{1}{Z_2}\end{bmatrix}$

$T=\begin{bmatrix}\frac{Z_2+Z_3}{Z_3} & \frac{Z_3(Z_1+Z_2)+Z_1Z_2}{Z_3}\\ \frac{1}{Z_3} & \frac{Z_3+Z_1}{Z_3}\end{bmatrix}$

B图 T型 (列方程或级联做)

① $\dot U_1=\dot I_1Z_1+(\dot I_1+\dot I_2)Z_3$

$\dot U_2=\dot I_2Z_2+(\dot I_1+\dot I_2)Z_3$

② $T_1=\begin{bmatrix}1 & Z_1\\ 0 & 1\end{bmatrix}$ $T_2=\begin{bmatrix}1 & 0\\ \frac{1}{Z_3} & 1\end{bmatrix}$ $T_3=\begin{bmatrix}1 & Z_2\\ 0 & 1\end{bmatrix}$

$T=T_1T_2T_3$ 用5-12

5-15

(1) $\dot U_1=Z_{11}\dot I_1+Z_{12}\dot I_2$ 可知 $Z=\begin{bmatrix}\frac{5}{3} & 3\\ -\frac{2}{3} & \frac{3}{2}\end{bmatrix}$

$\dot U_2=Z_{21}\dot I_1+Z_{22}\dot I_2$

$\begin{cases}\dot U_1=\frac{5}{3}\dot I_1+3\dot I_2\\ \dot U_2=-\frac{2}{3}\dot I_1+\frac{3}{2}\dot I_2\end{cases}\xrightarrow[\dot I_2=6]{\dot I_1=5}\begin{cases}\dot U_1=26.33V\\ \dot U_2=5.67V\end{cases}$

(2) $Y=\begin{bmatrix}-\frac{3}{2} & -6\\ \frac{1}{2} & -7\end{bmatrix}$ $\begin{cases}\dot I_1=-\frac{3}{2}\dot U_1-6\dot U_2\\ \dot I_2=\frac{1}{2}\dot U_1-7\dot U_2\end{cases}$

代入得 $\dot I_1=-7.5A$ $\dot I_2=-6.5A$

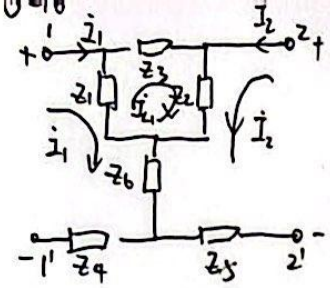
(3) $H=\begin{bmatrix}-3 & \frac{5}{3}\\ -2 & 0\end{bmatrix}$ $\begin{cases}\dot I_1=-3\dot U_1+\frac{5}{3}\dot I_2\\ \dot U_2=-2\dot U_1\end{cases}$

代入得 $\dot I_1=2.67A$ $\dot U_2=-6V$

(4) $T'=\begin{bmatrix}6 & -0.5\\ 5 & -0.3\end{bmatrix}$ $\begin{cases}\dot U_2=6\dot U_2-0.5(-\dot I_2)\\ \dot I_2=5\dot U_2-0.3(-\dot I_2)\end{cases}$

代入得 $\dot U_2=5.5V$ $\dot I_2=4.7A$

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利用网孔电流法

$\dot{U}_1 = -Z_1(\dot{I}_1-\dot{I}_{L1}) + Z_6(\dot{I}_1+\dot{I}_2) + Z_4\cdot\dot{I}_1$ ①

$\dot{U}_2 = Z_2(\dot{I}_{L1}+\dot{I}_2) + Z_6(\dot{I}_1+\dot{I}_2) + Z_5\dot{I}_2$ ②

$Z_1(\dot{I}_{L1}-\dot{I}_1) + Z_3\cdot\dot{I}_{L1} + Z_2(\dot{I}_{L1}+\dot{I}_2) = 0$ ③

⇒ 将 $\dot{I}_{L1}$ 替换掉.

由 ③ $\dot{I}_{L1} = \frac{Z_1}{Z_1+Z_2+Z_3}\dot{I}_1 - \frac{Z_2}{Z_1+Z_2+Z_3}\dot{I}_2$

故 $\dot{U}_1 = (Z_1+Z_4+Z_6-\frac{Z_1^2}{Z_1+Z_2+Z_3})\dot{I}_1 + (Z_6+\frac{Z_1Z_2}{Z_1+Z_2+Z_3})\dot{I}_2$

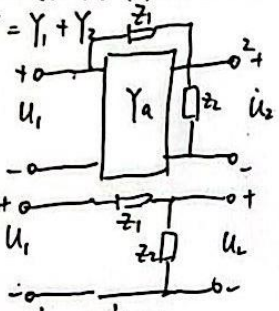
$\dot{U}_2 = (Z_6+\frac{Z_1Z_2}{Z_1+Z_2+Z_3})\dot{I}_1 + (Z_2+Z_5+Z_6-\frac{Z_2^2}{Z_1+Z_2+Z_3})\dot{I}_2$

$\therefore Z = \begin{bmatrix} Z_1+Z_4+Z_6-\frac{Z_1^2}{Z_1+Z_2+Z_3} & Z_6+\frac{Z_1Z_2}{Z_1+Z_2+Z_3} \\ Z_6+\frac{Z_1Z_2}{Z_1+Z_2+Z_3} & Z_2+Z_5+Z_6-\frac{Z_2^2}{Z_1+Z_2+Z_3} \end{bmatrix}$

5-17 用两双口网络的并联角度考虑

并联时 $Y = Y_1 + Y_2$

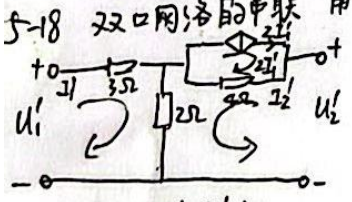
其中 Y_1 为

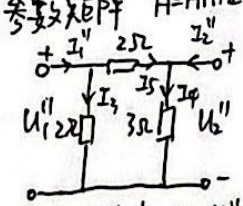


$Y_1 = \begin{bmatrix} \frac{1}{Z_1} & -\frac{1}{Z_1} \\ -\frac{1}{Z_1} & \frac{Z_1+Z_2}{Z_1Z_2} \end{bmatrix}$

$Y = Y_1 + Y_a = \begin{bmatrix} \frac{1}{Z_1}+Y_{a11} & Y_{a12}-\frac{1}{Z_1} \\ Y_{a21}-\frac{1}{Z_1} & Y_{22}+\frac{Z_1+Z_2}{Z_1Z_2} \end{bmatrix}$

5-18 22口网络的串联 用H参数矩阵 $H = H_1+H_2$





$\dot{U}_1' = 3\dot{I}_1' + 2(\dot{I}_1'+\dot{I}_2')$

$\dot{U}_2' = 4(2\dot{I}_1'+\dot{I}_2') + 2(\dot{I}_1'+\dot{I}_2')$

$\Rightarrow Z_1 = \begin{bmatrix} 5 & 2 \\ 10 & 6 \end{bmatrix}$

$H_1 = \begin{bmatrix} \frac{5}{3} & \frac{1}{3} \\ -\frac{5}{3} & \frac{1}{6} \end{bmatrix}$

$\dot{I}_3 = \dot{U}_1''\cdot\frac{1}{2}$ $\dot{I}_4 = \dot{U}_2''\cdot\frac{1}{3}$

$\dot{I}_5 = (\dot{U}_1''-\dot{U}_2'')\frac{1}{2}$

$\dot{I}_1'' = \dot{I}_3+\dot{I}_5$ $\dot{I}_2'' = \dot{I}_4-\dot{I}_5$

$\Rightarrow H_2 = \begin{bmatrix} 1 & \frac{1}{2} \\ -\frac{1}{2} & \frac{7}{12} \end{bmatrix}$

5-19

$\dot{U}_1 = (\dot{I}_1 - \frac{\dot{U}_1}{Z_a})\cdot Z_b + n\dot{U}_2$

化简 $\dot{U}_1 = -\frac{\dot{I}_2}{n}\cdot Z_b + n\dot{U}_2$

$\dot{I}_1 = \frac{\dot{U}_1}{Z_a} + \frac{\dot{U}_1 - n\dot{U}_2}{Z_b}$

$\dot{I}_1 = \frac{n}{Z_a}\dot{U}_2 + \frac{Z_a+Z_b}{nZ_a}(-\dot{I}_2)$

即 $T = \begin{bmatrix} n & \frac{Z_b}{n} \\ \frac{n}{Z_a} & \frac{Z_a+Z_b}{nZ_a} \end{bmatrix}$

5-20

$U_- = U_+ = 0$

$\frac{-U_2}{R\|\frac{1}{jwC_2}} = \frac{U_1}{\frac{1}{jwC_1}} \Rightarrow \frac{\dot{U}_2}{\dot{U}_1} = \frac{-RjwC_1}{jwC_2R+1}$

5-21

$\dot{I}_C = \frac{\dot{U}_1}{\frac{1}{jwC}+R_1}$ $\dot{U}_+ = \dot{I}_C\cdot R_1 = \frac{\dot{U}_1\cdot R_1}{\frac{1}{jwC}+R_1} = \dot{U}_-$

$\dot{I}_{R_2} = \frac{\dot{U}_1 - \frac{\dot{U}_1 R_1}{\frac{1}{jwC}+R_1}}{R_2} = \frac{\dot{U}_1}{(jwCR_1+1)R_2}$

$\dot{I}_1 = \dot{I}_C + \dot{I}_{R_2} = \dot{U}_1\left[\frac{jwC}{jwCR_1+1} + \frac{1}{(jwCR_1+1)R_2}\right]$

$\Rightarrow R_入 = \frac{\dot{U}_1}{\dot{I}_1} = \frac{(jwCR_1+1)R_2}{jwCR_2+1}$

5-22 用一系列虚短、虚断计算

$U_{1-} = U_{1+} = \dot{U}_1$ $I_1 = I_{R_3} = \frac{\dot{U}_2-\dot{U}_1}{R_3}$ 可惜 $\dot{U}_2$ 未知

$I_C = I_{R_1} = \frac{\dot{U}_1}{R_1}$ $U_{2-} = U_{2+} = \dot{U}_1$ 所以人 $U_C = U_{R_2}$

$U_{R_2} = U_C = I_C\cdot\frac{1}{jwC} = \frac{\dot{U}_1}{R_1\cdot jwC}$ $I_{R_2} = \frac{\dot{U}_1}{R_1R_2\cdot jwC}$

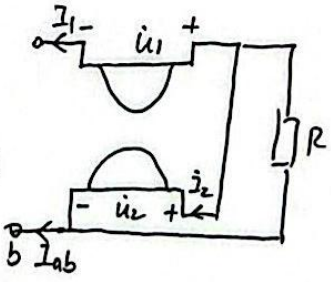
而 $U_{R_4} = U_{R_3} = I_{R_4}\cdot R_4 = I_{R_2}\cdot R_4 = \frac{\dot{U}_1\cdot R_4}{R_1R_2R_3jwC}$

$\dot{I}_1 = \dot{I}_{R_3} = \frac{\dot{U}_1\cdot R_4}{R_1R_2R_3\cdot jwC}$

5-23

$\begin{cases} U_{ab} = U_2 - U_1 \\ \dot{I}_{ab} = \dot{I}_2 + \frac{\dot{U}_2}{R} \\ \dot{U}_1 = -r\dot{I}_2 \quad \dot{I}_1 = -\dot{I} \\ \dot{U}_2 = -r\dot{I}_1 \end{cases}$

$\Rightarrow R_{ab} = \frac{\dot{U}_{ab}}{\dot{I}_{ab}} = \frac{r^2}{R}$



5-24 该电路为三个级联

$T_1 = \begin{bmatrix} 0 & \frac{1}{g} \\ g & 0 \end{bmatrix}$ $T_2 = \begin{bmatrix} 1 & 0 \\ jwC & 1 \end{bmatrix}$ $T_3 = \begin{bmatrix} 0 & \frac{1}{g} \\ g & 0 \end{bmatrix}$

$T = T_1T_2T_3 = \begin{bmatrix} 1 & \frac{jwC}{g^2} \\ 0 & 1 \end{bmatrix}$

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第六章（仅有6-4～6-15）（答题正做指：请在变换频率时标注此时有 X_C、X_L 值）

6-4

基波时 $i_0=0$ $P_0=0$

一次谐波时 $\dot{U}_1=100\angle 0°V$

$Z_{ab}=R+\frac{1}{j\omega C}+j\omega L=20\Omega$

$\dot{I}=\frac{\dot{U}}{Z_{ab}}=5\angle 0°A$

$P_1=UI\cos\varphi=500W$

三次谐波时

$\dot{U}_3=40\angle 30°V$

$Z_{ab}=R+j3\omega L+\frac{1}{j3\omega C}=20+j\frac{80}{3}\Omega$

$\dot{I}_3=\frac{\dot{U}}{Z_{ab}}=1.2\angle -23.13°$

$P_3=U_3I_3\cos\varphi=28.81W$

$i=i_1+i_3=5\sqrt{2}\sin\omega t+1.2\sqrt{2}\sin(3\omega t-23.13°)$

6-5 有效值的计算

$U=\sqrt{U_0^2+U_1^2+U_3^2}=50.5V$ $I=\sqrt{I_0^2+I_1^2+I_3^2}=5.92A$

$P=U_0I_0+U_1I_1\cos\varphi_1+U_3I_3\cos\varphi_3=182.94W$

6-6

①只有直流作用时

$I_{R(0)}=\frac{U_{(0)}}{R}=15A$

②一次谐波作用时

$Z_{ab}=j\omega L+R\|\frac{1}{j\omega C}=5\Omega$

$\dot{U}_1=100\sqrt{2}\angle 0°V$（注意不是200V，取有效值）

$\dot{I}_{L_1}=\dot{U}_1/Z_{ab}=20\angle 0°A$

$\dot{I}_R=\dot{I}_L\cdot\frac{j\omega C}{R+\frac{1}{j\omega C}}=20\angle -45°A$

$i_{R_1}=20\sqrt{2}\sin(\omega t-45°)$

③二次谐波作用时

$Z_{ab}=j2\omega L+R\|\frac{1}{j2\omega C}=6.32\angle 71.57°$

$\dot{U}_2=50\sqrt{2}\angle 90°V$ $\dot{I}_{L_2}=\frac{\dot{U}_2}{Z_{ab}}=5\sqrt{5}\angle 18.43°$

$\dot{I}_R=\dot{I}_{L_2}\cdot\frac{\frac{1}{2j\omega C}}{R+\frac{1}{2j\omega C}}=5\angle -45°A$

$i_{R_2}=5\sqrt{2}\sin(2\omega t-45°)$

$i_R=i_{R_0}+i_{R_1}+i_{R_2}=15+20\sqrt{2}\sin(\omega t-45°)+5\sqrt{2}\sin(2\omega t-45°)$

6-7（易错）

①基波作用时

$I_{1(0)}=\frac{U_{1(0)}}{2R}=0.2A$ ~~$U_{C0}=\frac{1}{2}U_{(0)}=20V$~~ × 被$L_2$短路

$u_{C0}=0V$

②一次谐波作用时

$Z_{ab}=(\frac{1}{j\omega C}\|j\omega L_2+j\omega L_1)\|R+R=2R=200\Omega$（谐振）

$\dot{I}_{(1)}=\frac{\dot{U}_{(1)}}{2R}=\frac{3}{40}\sqrt{2}\angle 0°A$ $\dot{U}_{C(1)}=\frac{1}{2}U_{(1)}=\frac{15}{2}\sqrt{2}\angle 0°V$

$i_{(1)}=0.15\sin\omega t(A)$ $u_{C(1)}=15\sin\omega t(V)$

③三次谐波作用时

$\dot{U}_{(3)}=35\sqrt{2}\angle 45°V$

$Z_{ab}=R+[(\frac{1}{j3\omega C}\|j3\omega L_2+j3\omega L_1)\|R]=200\Omega$

（串联谐振）

$i_3=\frac{u_3}{0.7R}=\frac{0.7}{0.7R}\sin(3\omega t+45°)$ $\dot{I}_3=0.35\sqrt{2}\angle 45°A$

~~$U_{C3}=0V$~~

$\dot{I}_C=\dot{I}_3\cdot\frac{j3\omega L_2}{\frac{1}{j3\omega C}+j3\omega L_2}=0.56\angle 45°$

$\dot{U}_C=\dot{I}_C\cdot\frac{1}{j\omega C}=18.56\angle -45°$

即 $u_C=18.56\sqrt{2}\sin(3\omega t-45°)$

$\therefore i(t)=[0.4+0.15\sin\omega t+0.7\sin(3\omega t+45°)]A$

$u_C(t)=15\sin\omega t+18.56\sqrt{2}\sin(3\omega t-45°)V$

6-8（用叠加定理做）

①当U_S单独作用时

$j\omega L=j20$ $\frac{1}{j\omega C}=-j20$

LC回路串联谐振

故R被短路 $i_R=0$

$U_S=ri_L\Rightarrow i_L(t)=\sqrt{2}\sin\omega t$ 故 $=i_C(t)$

②当I_S单独作用时

~~$j\omega L$~~ $j2\omega L=j40$ $\frac{1}{j2\omega C}=-j10$ $\dot{I}_S=3\angle 45°A$

有 $\dot{I}_L=\dot{I}_S+\dot{I}_C$

$\dot{I}_L\cdot j40+\dot{I}_C(-j10)+30\dot{I}_L=0$

$\Rightarrow\dot{I}_L=\frac{\sqrt{2}}{2}\angle -90°$ $\dot{I}_C=\frac{5\sqrt{2}}{2}\angle -126.87°$ $\dot{I}_R=\frac{3\sqrt{2}}{2}\angle -90°$

即 $i_R=3\sin(2\omega t-90°)$ $i_C=\sqrt{2}\sin\omega t+5\sin(2\omega t-126.8°)$

$i_L=\sqrt{2}\sin\omega t+\sin(2\omega t-90°)$

U_S单独作用时

$\dot{I}'_{U_S}=I_C-I_R=\dot{I}_L=1\angle 0°A$ $\dot{U}_S=30\angle 0°$

$\dot{U}'_L=\dot{I}_L\cdot j\omega L=20\angle 90°V$

$U'_{iS}=20\angle 90°V+ri_L=10\sqrt{13}\angle 33.69°V$

I_S单独作用时 ~~$I_S=3$~~

$\dot{I}''_{U_S}=\dot{I}_C-\dot{I}_R=2.24\angle -161.57°$

$\dot{U}''_L=\dot{I}'_L\cdot j\omega L=20\sqrt{2}\angle 0°$

$U''_{iS}=\dot{U}''_L+ri''_L=25\sqrt{2}\angle -36.87°$

~~$P=P_1+P_2$~~ $I_S=3\angle 45°$

$P_{IS}=15W$ $P_{US}=30W$

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6-9

① U_2单独作用时

$i_1'=-1A \quad i_2=1A \quad i_3=0$

$P_{u_2}=U_2\cdot I_2=100W$

② U_1单独作用时

$Z=j\omega L_1+j\omega L_2 // \frac{1}{j\omega C}+R=126.6\angle 37.82°$

$\dot{I}_1=\frac{220\angle 0°}{Z}=1.738\angle -37.82°$

$\dot{I}_2=-\dot{I}_1\cdot\frac{\frac{1}{j\omega C}}{j\omega L_2+\frac{1}{j\omega C}} \quad \dot{I}_3=\dot{I}_1\cdot\frac{j\omega L_2}{j\omega L_2+\frac{1}{j\omega C}}$

$\dot{I}_2=3.427\angle 142.18° \quad \dot{I}_3=1.69\angle 142°$

$P=U\cdot I\cos\varphi=220\times 1.738\cos 37.82°=302.04W$

$i_1(t)=-1+1.738\sqrt{2}\sin(\omega t-37.82°)$

$i_2(t)=1+3.427\sqrt{2}\sin(\omega t+142.18°)$

$i_3(t)=1.69\sqrt{2}\sin(\omega t+142°)$

6-10

① U_S单独作用时 串联谐振 注意要代入100

$I_{R1}=\frac{U_S}{R}=5\sqrt{2}A \quad P_{U_S}=\frac{150\sqrt{2}}{\sqrt{2}}\times 5\sqrt{2}=500W$

② I_S单独作用时

$U_{IS}=R_2\cdot I_S=10V \quad P_{I_S}=10\times 1=10W$

6-11

$\omega=2rad/s$时 C与L_2并联谐振

$\frac{1}{\omega C}=\omega L_2 \Rightarrow L_2=1H$

$\omega=4rad/s$时 C和L_2与L_1串联谐振

$j\omega L_1+\frac{1}{j\omega C}// j\omega L_2=0 \Rightarrow L_1=\frac{1}{3}H$

利于$P_{总}$计算R

~~基波~~直流时 ~~干~~ $P_1=\frac{10^2}{R}=\frac{100}{R}$（有一个R被短路）

$\omega=2$时 $P_2=\frac{20^2}{2R}=\frac{200}{R}$

$\omega=4$时 $P_3=\frac{10^2}{R}=\frac{100}{R}$

$\frac{100}{R}+\frac{200}{R}+\frac{100}{R}=100 \Rightarrow R=4\Omega$

6-12

仅直流 $i_L(t)=\frac{U_0(t)}{R_1+R_2} \quad U_0=200V$

仅基波

$\dot{U}_C=\dot{I}_L(j\omega L+R_2)=(100+j100)V$

$\dot{I}_C=\frac{\dot{U}_C}{\frac{1}{j\omega C}}=\sqrt{2}\angle 135°$

$\dot{I}_{R_1}=\dot{I}_C+\dot{I}_L=1\angle 9°$

$\dot{Z}=R_1+\frac{1}{j\omega C_1}// (j\omega L+R_2)=223.61\angle 26.57°$

$\dot{U}=\dot{I}_{R_1}\cdot Z=223.61\angle 63.43°$

仅三次谐波作用时

$\dot{I}=\dot{I}_L+\dot{I}_C=4.27\angle 139.44°$

$Z_{ab}=R_1+\frac{1}{j3\omega C}// (j3\omega L+R_2)=107.91\angle -20.05°$

$U=\dot{I}\cdot Z_{ab}=460.76\angle 139.39°$

$U=\sqrt{U_{(0)}^2+U_{(1)}^2+U_{(3)}^2}=\sqrt{220^2+223.61^2+460.76^2}=557.41V$

6-13 五次谐波为负序系统，三次谐波为零序系统

基波为正序系统

① 三次谐波

$\dot{I}_{A3}=\frac{U_{A3}}{3Z_0+Z_L}=\frac{\frac{1}{4}\times\frac{220}{\sqrt{2}}}{3(10+j\frac{1}{3})+50+j50}=0.0798\angle -66.04°$

$\dot{I}_0=\sqrt{3}\dot{I}_{A3}=0.239\angle -66.04°$

② 一次谐波

$\dot{U}_{NN(1)}=\frac{\frac{\dot{U}_{A1}}{Z_L}+\frac{\dot{U}_{B1}}{Z_L}+\frac{\dot{U}_{C1}}{Z_L}}{\frac{3}{Z_L}+\frac{1}{Z_0}}=0$

$\dot{U}_{A_1}=100\sqrt{2}\angle 0° \quad \dot{U}_{B_1}=100\sqrt{2}\angle -120° \quad \dot{U}_{C_1}=100\sqrt{2}\angle 120°$

$\dot{I}_{A(1)}=\frac{100\sqrt{2}\angle 0°}{50+j50}=2\angle -45°A$

③ 五次谐波

$\dot{I}_{A(5)}=\frac{4\sqrt{2}\angle 0°}{50+j250}=0.022\angle -78.7°$

$i_A(t)=2\sqrt{2}\sin(\omega t-45°)+0.11\sqrt{2}\sin(3\omega t-66.04°)+0.031\sin(5\omega t-78.7°)$

$i_0(t)=0.239\sqrt{2}\sin(3\omega_1 t-66.04°)$

6-14 无中性线 ⇒ 相（线）（电压/电流）不包含零序分量

$u_{AB}=\sqrt{2}\times 90\sin\omega t+\sqrt{2}\times 50\sin(5\omega t+90°)$

$u_{AB}=\sqrt{6}\times 90\sin(\omega t-30°)+\sqrt{6}\times 50\sin(5\omega t+60°)$ [一个为正序，一个为负序]

$U_{(A)}=\sqrt{90^2+50^2}=102.96V \quad U_{(AB)}=\sqrt{(\sqrt{6}\times\frac{90}{\sqrt{2}})^2+(\sqrt{6}\times\frac{50}{\sqrt{2}})^2}=178.31$

$\dot{I}_{A(1)}=\frac{90\angle 0°}{j6-j30}=3.75\angle 90°A$

$\dot{I}_{A(5)}=\frac{50\angle 90°}{j30-j6}=2.08\angle 0°$

$I_A=\sqrt{3.75^2+2.08^2}=4.29A$

6-15

$220=\sqrt{U_{(1)}^2+U_{(3)}^2+U_{(5)}^2}=\sqrt{U_1^2+(\frac{1}{3}U_1)^2+(0.15U_1)^2}$

$\Rightarrow U_{(1)}=213.93V \quad U_{(3)}=42.69V \quad U_{(5)}=32.01V$

三次谐波零序系统 无线电压

故 $U_{线}=\sqrt{(U_{(1)}\cdot\sqrt{3})^2+(U_{(5)}\cdot\sqrt{3})^2}=373.81V$

$I_{线}=I_{相}=\frac{U_{相}}{Z}=2.2A$

中性线上仅有三次谐波产生的电流

$I_0=\frac{42.69}{100}\times 3=1.28A$

三相产生电流相同